

# VIX Maturity Interpolation\*

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## Abstract

The VIX index is computed as a weighted average of two model-free option-implied variance measures for maturities around 30 days. This induces an interpolation error which, since 2014, could be sizable and especially pertinent under stressed market conditions. We quantify the VIX interpolation errors and identify underlying sources. We show the change to include weekly options for the VIX calculation in 2014 led to a tenfold error reduction, eliminating them as a practical concern. Thus, the gap between the old and new VIX measure is a good proxy for the interpolation error. Moreover, the new Cboe VIX calculation is not backdated, so the series displays different statistical properties before and after 2014.

Our analysis is relevant for a broad set of VIX-style indices worldwide, since nearly all such measures are computed exclusively from monthly options. In addition, there is growing interest in shorter-horizon VIX indices, and for such measures the interpolation errors amplify notably. Our analysis facilitates assessment of these errors and offers guidance for new robust computational algorithms. We exemplify this via an analysis of the 9-day VIX index.

Finally, we compare the interpolation scheme for VIX to a pair of existing alternatives also based on monthly options. We find that such modifications can reduce the average errors, but they are neither uniformly superior nor do they resolve the basic issue satisfactorily.

**The views expressed in this article are those of the authors and do not necessarily coincide with those of the Banco de España and the Eurosystem.**

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# 1 Introduction

The model-free volatility index, VIX, was introduced at the Chicago Board Options Exchange (CBOE, now Cboe Global Markets, and henceforth referred to as Cboe) in 2003, as replacement for an existing index (the old VIX) constructed from Black-Scholes implied volatilities for at-the-money (ATM) options on the S&P 100 equity index. The new model-free VIX methodology relies on a broad cross-section of out-of-the-money (OTM) SPX put and call options, written on the S&P 500 equity index, with tenors around 30 calendar days. The new volatility series was subsequently backdated until 1990, so a model-free VIX series covering more than thirty years is now available.<sup>1</sup> The analytical tractability of the new VIX measure facilitates both pricing and hedging, and Cboe introduced VIX futures in 2004 and VIX options in 2006. These volatility derivatives rapidly gained traction and are among the most successful product innovations in the history of exchange-traded securities. The same calculation formula is now used by the Cboe for volatility measurement across a range of other calendar horizons and for a variety of asset classes. Moreover, the methodology has been licensed for use on equity indices in a number of foreign countries, while closely related approaches are adopted by other leading global exchanges.<sup>2</sup>

A less appreciated, but noteworthy, change in methodology occurred on October 6, 2014. On that date, Cboe modified the calculation of the VIX to include SPX weekly options. Previously, the VIX relied exclusively on monthly or quarterly SPX options. The “weeklys” allow for a closer match to the target maturity of 30 calendar days, as only options with tenors between 23 and 37 days are used, in contrast to maturities ranging from 7 to 67 days for monthly options. Following this change, the official Cboe VIX index refers to the computation involving weekly options. Until recently, the prior methodology, based exclusively on monthly options, continued to be disseminated under the ticker VIXMO (VIX Month Only). Since reliable quotes for weekly SPX options are not available going back in time, the series was not backfilled. Thus, the historical VIX series is identical to VIXMO for the majority of the sample, and switches to the new methodology involving weekly options only in recent years. For clarity, we refer to the latter measure as VIXWE, whenever we need to distinguish the two computational procedures. In summary, the existing official VIX series is a mixture of two indices, consisting of VIXMO prior to October 6, 2014, and VIXWE thereafter.

Of course, VIXMO and VIXWE are designed to measure the same quantity, the Model-Free option-Implied return Volatility (MFIV) for the S&P 500 index over the coming 30 calendar days. Hence, the decision to change the computational procedure was motivated solely by a perceived improvement in precision. First, the inclusion of weeklys ensures that the option maturities better match the target of 30 days. Second, with tenors between 23 and 37 days, the VIXWE value always represent an interpolation of two maturities along the volatility term structure, while the prior procedure involved extrapolation at some stages within the monthly maturity cycle. This shift in methodology has passed without much commentary from non-exchange observers, academics, or users, reflecting the perception that the index is largely unaltered, although perhaps now a bit more robust due to a reduction in noise.

We document that, in fact, VIXWE deviates systematically from VIXMO, and occasionally non-trivially so. Figure 1 depicts the ratio of VIXMO to VIXWE from October 6, 2014, to December 2020. The ratio has a mean of 1.0099 (1.0097), a minimum value of 0.900 and a maximum value of 1.316 (1.098), where the figures in parentheses reflect the exclusion of a single

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<sup>1</sup>Until recently, a version of the old VIX measure, based on the S&P 100 index, continued to be calculated and released under the ticker symbol VXO.

<sup>2</sup>For example, the methodology for computing the New-VDAX volatility index at the Deutsche Börse is almost identical to the one for VIX, although the filtering rules for inclusion of OTM options are different.

outlier in July 2020. Hence, we observe discrepancies of up to 10%, while the mean absolute deviation (from unity), MAD, equals 0.0197 (0.0196). In summary, there are substantial relative deviations on both the upside and downside. Moreover, the divergences can change sign over the course of a few trading days, implying that abrupt shifts in the ratio of 15% are not uncommon.

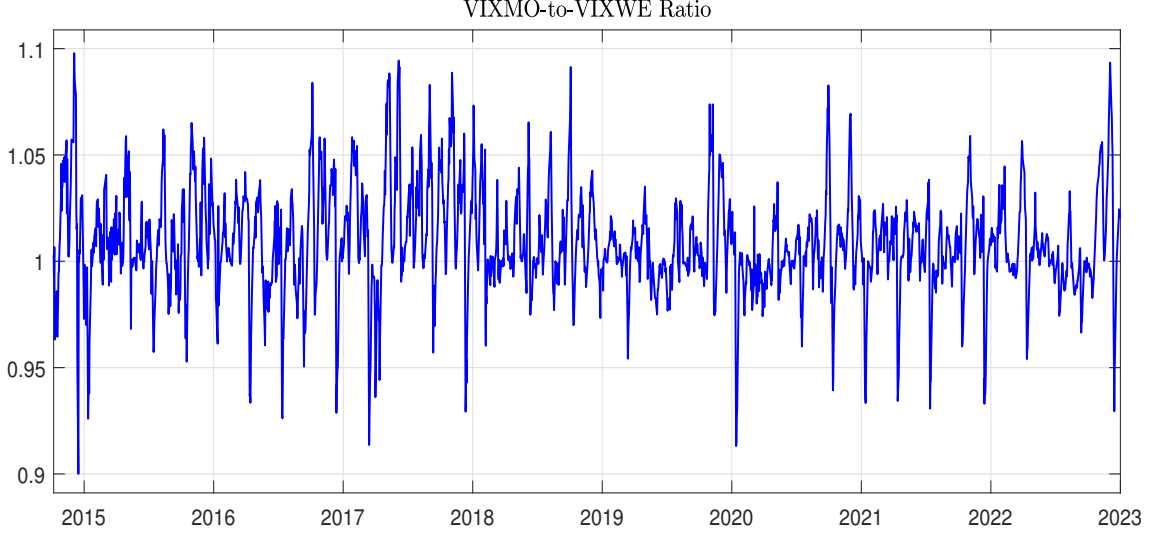


Figure 1: The ratio of VIXMO-to-VIXWE depicted over time.

The fluctuations in Figure 1 may appear random, but they embed a highly systematic component. Figure 2 reveals that the VIXMO/VIXWE ratio displays a very strong cyclical pattern, with a main period of around 21 trading days, or one calendar month, while another distinct spike is visible around every 63 trading days, or 3 calendar months.

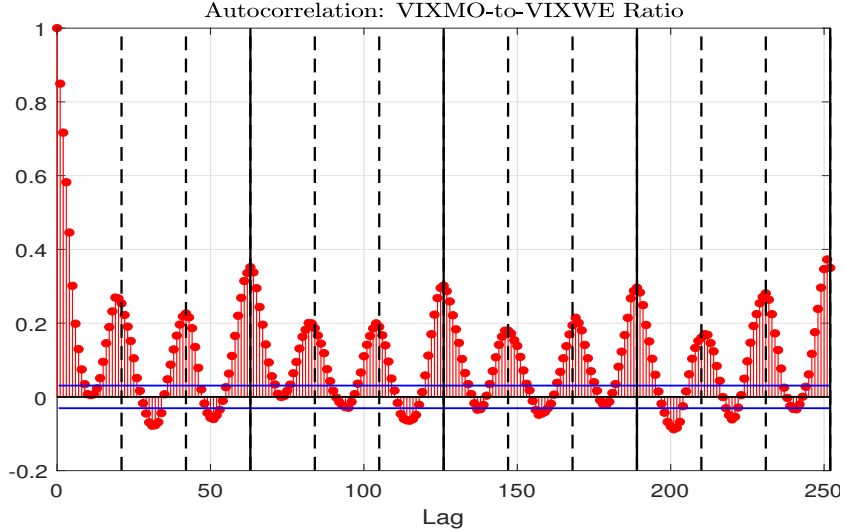


Figure 2: Autocorrelogram for the VIXMO-to-VIXWE ratio. The horizontal axis indicates the lag length measured in exchange trading days.

This finding raises a number of questions. First, can we corroborate and quantify the extent to which the shift from VIXMO to VIXWE lowers the interpolation error? Second, is the

systematic deviation between VIXMO and VIXWE due solely to a mechanical change in the maturity structure of the underlying options, or do other factors contribute substantively to these discrepancies? Third, what are the advantages and drawbacks of alternative linear interpolation schemes? Finally, will the wider universe of volatility indices benefit from similar design changes?

Our paper addresses these issues. At a broader level, we fill a gap in the literature concerning the source and extent of measurement errors in option-implied model-free volatility indices. Jiang & Tian (2007) classifies them as arising from the following types of approximation: (i) truncation errors due to the use of a finite strike range; (ii) discretization errors arising from approximating an integral over the strike range with a discrete sum based on the discrete set of exchange-listed strikes; (iii) a Taylor series expansion approximation errors for the log-function around the ATM strike; and (iv) linear maturity interpolation errors associated with the use of VIX values for neighboring maturities to generate the 30-day interpolated value.

Jiang & Tian (2007) analyze the errors associated with the first three approximations through simulations with different data generating processes. Notably, however, they decline to assess the maturity interpolation error, merely noting that this error is linked to the degree of non-linearity of the volatility term structure, which varies substantially over time. Since monthly options, by construction, have a coarse grid of tenors, the curvature of the term structure at the 30-day maturity is not observable, but must be approximated through the imposition of strong assumptions. Hence, the paucity of *monthly* options with maturities around 30 days and the declining liquidity for longer-dated options render assessment of any specific approach to maturity interpolation speculative in nature. As a result, the VIX maturity interpolation error has never been seriously addressed nor quantified in the extant literature.

We document that the use of *weekly* options in the VIX computation dramatically lessens the problems surrounding maturity interpolation, allowing us to address the implications of the shift from VIXMO to VIXWE directly. Specifically, we show that the interpolation errors associated with VIXWE are negligible and, thus, the deviation between the two measures provides a direct indication of the VIX interpolation error due to reliance on monthly options (VIXMO). Hence, to a very good approximation, the deviations from unity of the VIXMO-to-VIXWE ratio in Figure 1 are an accurate measure of the relative interpolation errors of the traditional VIX measure. Since we have observations for both VIXMO and VIXWE from 2014 through 2023, we now have a substantial sample size from which to assess the resulting errors.

Our results are based on SPX options and do not provide direct evidence about the maturity interpolation errors for other VIX-style measures. Nonetheless, since we derive a set of approximation formulas that help identify the main drivers behind these errors, we may assess them more generally as a function of the option maturity structure, horizon, and, inevitably, the shape of the volatility term structure. This enables us to speak to the interpolation errors for other option-implied measures computed for numerous national equity indices, individual stocks, fixed income securities, and commodities, along with different target maturities, such as 9 days, 3 months, and one year. And, perhaps most importantly, the CME Group has recently been rolling out a suite of MFIV indices, labeled CVOL, using a different maturity interpolation technique. We refer to this approach as “annualized variance” rather than the usual “total variance” interpolation used by all other prominent MFIV measures. In our analysis, we also explore the implications and relative benefits of using either of these interpolation procedures.

The remainder of the article is organized as follows. Section 2 reviews the VIX-style measures disseminated worldwide. Section 3 analyzes the maturity interpolation technique used by Cboe and contrasts it with a couple of alternatives, popular among academics and adopted by a few exchanges. We determine the main drivers behind the errors and introduce a convenient measure capturing the impact of the maturity structure, labeled the Convexity Adjustment

Factor, or CAF. We also explore the biases within a calibrated version of the Heston Stochastic Volatility (SV) model. Section 4 presents our empirical results based on monthly and weekly SPX options. Section 4.1 starts with an overview of our SPX option sample. The following sections then provide illustrative examples of interpolation errors and quantify their impact, highlighting the effect of the underlying option maturity structure. Section 4.4 uses our newly developed historical simulation technique to greatly expand the empirical evidence by generating the full array of maturity structure each trading day using the rich set of weekly options. Further exploiting this approach, Section 4.5 documents the significant impact of the volatility term structure. Finally, Section 5 demonstrates that interpolation errors present an even larger issue for the short-maturity 9-day VIX, also disseminated by the Cboe, and we recommend a practical approach to greatly mitigate this problem. Section 6 provides concluding remarks.

## 2 The Dissemination of VIX-Style Measures

The most prominent option-implied volatility index is unquestionably the 30-day annualized VIX measure for the S&P 500 index disseminated by the Cboe. It serves as the underlying index for VIX derivatives, notably futures and options, and as a volatility or uncertainty gauge in academic work. Meanwhile, the popular press widely cites it as the market’s “fear gauge.”

### 2.1 The Development of Cboe Volatility Derivatives

The procedures behind the computation of the Cboe VIX index have evolved substantially over time. Table 1 provides a timeline for notable developments regarding the underlying S&P (SPX) options, the VIX methodology itself, and the VIX derivatives markets. For the main series of interest, we also provide the current ticker symbol (in bold). Notice that tickers also are subject to changes, so some series were disseminated under a different ticker symbol in the past.

The innovations outlined in Table 1 fall into three categories. The first group consists of changes in the set of option prices used. The original VIX measure, introduced in 1993, and backfilled to 1986, relied on monthly ATM S&P 100 options. However, from 2003, the VIX uses both OTM and ATM S&P 500 monthly options, backfilled until 1990. Finally, since 2014, the VIX uses OTM and ATM S&P 500 weekly (Friday expiration) and monthly options. The second group refers to shifts in the computational procedure, with the 2003 modification representing a move from a model-based to a model-free measure. Finally, the third group refers to the inception of trading in SPX derivatives (text in red) and VIX derivatives (text in blue).

Our primary focus is on one specific modification in the VIX calculation – the inclusion of Friday weekly SPX options in the index calculation since 2014. The critical difference relative to prior computational innovations is that the information set reflected in the VIX index was modified, but those changes were not – and could not be – backfilled, as weekly options were unavailable beyond the recent past. Thus, this constitutes a structural change, which may render the VIX series heterogenous across time, consisting of VIXMO prior to the shift in 2014 and VIXWE thereafter. This modification has not drawn much attention or been subject to careful analysis. This probably reflects the impression that the change is innocuous, largely representing a reduction of idiosyncratic noise in the index due to a mitigation of the interpolation error.

From a scholarly perspective, however, the computational shift is analogous to a natural experiment, allowing us to gauge the historical precision of the VIX index. We document that, for practical purposes, the new procedure renders the interpolation error negligible. Therefore, deviations between VIX computed from the new and old set of option tenors identify the interpolation error. Prior to the emergence of options spanning a dense set of expiries around the

Table 1: Timeline for major events affecting SPX options, the VIX methodology, and the trading of VIX options and futures. Text in red (blue) refers to SPX (VIX) options.

|      |   |                                                                                                                                                                                                                          |
|------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1993 | • | January 19: VIX launched (ticker: <b>VXO</b> ). ATM IV for S&P 100 monthly options. Series available for 1986-2021. Discontinued on September 23, 2021                                                                   |
| 2003 | • | September 22: VIX becomes model-free (ticker: <b>VIXMO</b> ). ATM and OTM S&P 500 monthly options. AM-settled. Series available for 1990-today                                                                           |
| 2004 | • | March 26: <b>VIX futures start trading</b> (ticker: <b>VX</b> )                                                                                                                                                          |
| 2005 | • | <b>SPX weekly options are listed</b> . Friday, PM-settled (ticker: <b>SPXW EOW</b> ).                                                                                                                                    |
| 2006 | • | <b>VIX options are listed</b> (ticker: <b>VIX</b> options)                                                                                                                                                               |
| 2010 | • | <b>VIX weekly options are listed</b> (ticker: <b>VIXW</b> options)                                                                                                                                                       |
| 2012 | • | <b>Expansion of listings for SPX weekly options (Weeklys)</b> . Minimum of five consecutive expiration weeks available for trading S&P 500 options                                                                       |
| 2014 | • | October 6: VIX computed from monthly and Friday weekly SPX options (ticker: <b>VIX</b> ). ATM, OTM S&P 500 options. Monthly options (1990-2014) and monthly/weekly options (2014-today). Merged series covers 1990-today |
| 2015 | • | <b>VIX weekly futures are launched</b> (ticker: <b>VXW</b> )                                                                                                                                                             |
| 2016 | • | February 23: <b>SPXW Weeklys, Wednesday expiry</b> (ticker: <b>SPXW (WED)</b> )                                                                                                                                          |
| 2016 | • | August 15: <b>SPXW Weeklys, Monday expiry</b> (ticker: <b>SPXW (MON)</b> )                                                                                                                                               |
| 2020 | • | August 9: <b>Mini VIX futures launched</b> (ticker: <b>VXM</b> )                                                                                                                                                         |
| 2022 | • | April 18: <b>SPXW Weeklys with Tuesday expiry</b> (ticker: <b>SPXW (TUE)</b> )                                                                                                                                           |
| 2022 | • | May 11: <b>SPXW Weeklys with Thursday expiry</b> (ticker: <b>SPXW (THU)</b> )                                                                                                                                            |

30-day horizon, this type of model-free analysis was infeasible. Hence, the methodological tweak in 2014 provides us with an opportunity to explore the nature of the interpolation bias, which deliberately was left unexplored in the Jiang & Tian (2007) study.

## 2.2 Alternative VIX-Style Indices

We focus on the Cboe 30-day VIX index, given the outsized attention it commands worldwide, and the role it serves as the underlying for the large and highly liquid VIX derivatives markets. However, it is also of interest to understand to what extent any lessons gained from the Cboe VIX index apply across the broader universe of volatility indices.

Among global exchanges, the Cboe and CME Group disseminate the most comprehensive set of U.S. option-implied volatility indices, covering various asset classes and horizons. Internationally, the most prominent volatility measures cover national or continent-wide equity indices. Table 2 summarizes key features of the maturity interpolation methodology applied for a set of the most important indices. For conciseness, we report only measures for the implied one-month risk-neutral volatility, although shorter and longer horizons are available in some cases. Moreover, outside of the U.S., we focus on equity volatility indices, as the remainder are less widely followed, and we often lack detailed information regarding their construction.

The table lists the maturity cycle for the underlying options; the index label; the minimum option tenor admissible for index computation, T; the quoting exchange; and whether the volatility measure serves as an underlying index for active derivatives trading. The dominant section of the table refers to indices that approximate the variance at the 30-day target horizon

by linearly interpolating or extrapolating the option-implied measures from nearby maturities quoted in terms of “total variance.” In contrast, the bottom part of Table 2 refers to the procedures for the CVOL index calculation at the CME Group, which rely on linear interpolation of option-implied measures quoted in terms of “average variance.” The distinction between “total” and “average” variance is discussed at length in Section 3.<sup>3</sup>

The Cboe, MIAAX, and CME Group are the main U.S. exchanges disseminating volatility indices. In particular, the Cboe computes model-free volatility indices for equity indices (S&P 500, Nasdaq 100, Dow Jones 30, and Russell 2000), individual stocks (Apple, Amazon, Google, IBM, and Goldman Sachs), the euro currency, and international equity ETFs covering broader non-North American markets, emerging markets, Brazil, and China, and MSCI indices, along with U.S. Treasuries and commodities. Among these, the 30-day VIX relies on weekly SPX options with a minimum of 23 days to maturity, alongside the RVX, VXN and other volatility indices referring to specific stocks, e.g., VXAPL, VXAZN, VGOG, VGS, and VIXIBM. However, some non-VIX Cboe volatility indices still rely on monthly options with a minimum of seven days to expiration.

The MIAAX Exchange Group calculates the SPIKES volatility index for the S&P 500 equity index using SPY options. Similar to VIX, SPIKES seeks to capture S&P 500 MFIV over the coming 30 calendar days, but there are non-trivial differences between the two methodologies; see Carr & Figà-Talamanga (2021). In particular, SPIKES relies exclusively on monthly options, so it resembles VIXMO rather than VIXWE in maturity interpolation. Moreover, SPIKES allows for the use of options with as little as two days to maturity, while the Cboe VIXMO required monthly options to have at least seven days to expiration, and the current VIXWE uses options with at least 23 days to maturity. These differences in the maturity cycle induce distinct roll-over effects and associated interpolation errors, as demonstrated later.

The CME Group has developed a suite of CVOL indices based on futures options. They do not cover equity indices, but most other major asset classes, including fixed income and foreign exchange markets, energy, agriculture, and metals.<sup>4</sup> Interestingly, the CME Group relies on different option tenors for the index computation depending on the available maturity structure of the underlying options. Thus, depending on the index, the CVOL maturity interpolation procedure resembles either VIXMO or VIXWE. As noted previously, CVOL also adopts the notion of average variance (AV) rather than total variance (TV), resulting in a modified weighting of the options in the computation of the volatility measure. We explore the relative advantages of the TV versus AV approach for maturity interpolation in Section 3.

S&P collaborates with the Toronto Stock Exchange to compile the one-month MFIV index for the S&P/TSX 60 stock ETF based on monthly options for the Canadian equity market, so it mimics the VIXMO interpolation approach, with only minor computational differences.

The introduction of volatility indices in Europe has been less successful. STOXX provides two volatility measures for equity indices, that currently serve as underlying for derivatives trading, namely VDAX-NEW for the German DAX and VSTOXX for the Eurostoxx 50 index. Both measures rely on monthly options, with near-term options rolling over very close to the maturity date. A few volatility indices are disseminated despite the absence of associated derivatives

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<sup>3</sup>Another major distinction between CVOL and the other indices is the former’s use of the so-called simple variance methodology. The basic insight behind this approach was provided by Carr & Corso (2001), and it constitutes a special case of the generalized variance contracts studied by Bondarenko (2014). It is also the index construction underlying the analysis of Martin (2017). Because both the regular and simple VIX face the identical interpolation issues, we restrict our attention to the standard (Cboe) MFIV measure.

<sup>4</sup><https://www.cmegroup.com/market-data/cme-group-benchmark-administration/files/cvol-methodology.pdf> provides additional details

Table 2: **Interpolation conventions for option-implied volatility measures.** OMC signifies the “Option maturity cycle,” with (W) indicating weekly and (M) monthly options. The minimum tenor (in calendar days) for the near-maturity option to be used in the volatility index construction is denoted  $\underline{T}$ . The computation rolls from the nearby to next maturity option, if  $T_1 < \underline{T}$  otherwise. Deriv indicates whether derivatives are traded on the volatility index (VI). Note: (\*) Derivatives are available only for some volatility indices.

### Total Variance Interpolation

| OMC | Volatility Indices (1M)                      | $\underline{T}$ | Exchange         | Deriv |
|-----|----------------------------------------------|-----------------|------------------|-------|
| W/M | VIX (S&P 500, Nasdaq, Russell, stocks, MSCI) | 23              | Cboe             | Yes   |
| M   | VIX (Currency, Fixed-Income, Commodities)    | 7               | Cboe             | Yes*  |
|     | SPIKES (U.S.)                                | 2               | MIAX             | Yes   |
|     | S&P/TSX 60 VIX (Canada)                      | 6               | TSX              | No    |
|     | VSMI (Switzerland)                           | 2               | SIX              | No    |
|     | VDAX-NEW, VSTOXX (Germany, Eurozone)         | 2               | STOXX Ltd.       | Yes   |
|     | FTSE 100 IVI, FTSE MIB IVI (UK, Italy)       | 7               | FTSE Russell     | No    |
|     | HSI VI (Hong Kong)                           | 4               | HSI Company Ltd. | Yes   |
|     | Nikkei 225 VI (Japan)                        | 4               | Nikkei Inc.      | Yes   |
|     | V-KOSPI 200 (South Korea)                    | n.a.            | KRX Market       | Yes   |
|     | TAIWAN VIX (Taiwan)                          | 3               | Taifex           | No    |
|     | NVIX (India)                                 | 4               | NSE              | No    |
|     | S&P/ASX 200 VIX (Australia)                  | 7               | ASX Ltd.         | No    |
|     | RVI (Russia)                                 | 8               | MOEX MICEX-RTS   | No    |
|     | VIBEX (Spain)                                | 1               | BME              | No    |

### Annualized Variance Interpolation

| OMC | Volatility Indices (1M)                             | $\underline{T}$ | Exchange  | Deriv |
|-----|-----------------------------------------------------|-----------------|-----------|-------|
| W/M | CVOL (Fixed-Income, FX, metal, energy, agriculture) | 10/14           | CME Group | No    |



trading. FTSE Russell distributes a volatility index for the U.K. (FTSE IVI) and Italy (FTSE MIB IVI), while the SIX (Swiss Stock Exchange) disseminates the VSMI for the Swiss SMI. Finally, Euronext used to publish volatility indices for AEX (the Netherlands), FTSE 100 (United Kingdom), CAC (France), and BEL (Belgium) markets, but they have since been delisted.

The Russian stock market (MOEX) releases the RVI, a volatility index from near and far-term ATM and OTM RTS index options with more than seven days to maturity. The Hang Seng Indexes Company Ltd. computes a volatility index from HSI options around the target maturity, rolling to new tenors four days prior to expiry. Nikkei Inc. estimates a Japanese volatility index (VNKY) from Nikkei 225 futures and options on the Osaka Exchange, rolling on to longer nearby tenors three business days before expiration. Osaka University provides an alternative one-month volatility index for the Nikkei 225 stock index with a roll-over analogous to VIXMO, while S&P Global computes a 10-year Japanese Bond volatility index (S&P/JPX JGB). Taifex releases a volatility index for the Taiex stock index using options with a maturity in excess of two days (TAIWAN VIX). The National Stock Exchange of India calculates the NVIX following the Cboe methodology, using NIFTY options with tenors beyond three days. The South Korea Stock Exchange releases the Kospi 200 MFIV (V-KOSPI 200), but little information is available regarding its specific variance interpolation and maturity rollover procedures. Finally, the S&P/ASX 200 VIX is a volatility index that seeks to measure the 30-day implied volatility in the Australian stock market using near and far-term ASX 200 options with seven or more days to maturity, following the VIXMO minimum option maturity criterion.

There are also examples of exchanges not adhering to the model-free methodology, likely due to the lack of liquidity of the OTM options. In this category, the S&P/BMC IPC VIX measures the 90-days IPC implied volatility for the Mexican stock market using options with a minimum of ten days to expiration. The Johannesburg Stock Exchange distributes a 90-days volatility index for the SAVI Top 40 equity index as the at-the-money volatility adjusted for the volatility skew. Using a similar methodology, the BME (Six-Group) computes a one-month volatility index for the Spanish stock index Ibex-35, the VIBEX. In all cases, the ATM option prices are assigned higher weights than those for OTM options.

The Cboe has expanded its footprint in European<sup>5</sup> and Asian markets<sup>6</sup> significantly over the last years. As part of this process, the Cboe has introduced their own version of existing national and continent-wide equity indices in Europe (see <https://www.cboe.com/europe/indices/>), thus setting the stage for the introduction of derivatives trading. In fact, Cboe Europe Derivatives (CEDX)<sup>7</sup> already issued futures and options on Cboe Germany 40 Index (September 21, 2021), and the Cboe Eurozone 50, France 40, Netherlands 25, Switzerland 20 and UK 100 Indices (September 6, 2021),<sup>8</sup> and equity options.<sup>9</sup> The advantages of a homogeneous set of national indices trading within a single uniform marketplace could well spur improved transaction volumes and liquidity, creating a foundation for successful VIX style markets covering all prominent European equity indices. The move into Asia is more recent, but may well foreshadow similar developments for that continent. Our analysis of alternative MFIV computational schemes may inspire the adoption of design features - both for existing and newly introduced volatility indices - that mitigate the sensitivity of MFIV indices to systematic maturity interpolation errors.

<sup>5</sup>Visit the Cboe Europe Equity for further details at <https://www.cboe.com/europe/indices/>

<sup>6</sup>For further information, see <https://ir.cboe.com/news-and-events/2021/06-30-2021/cboe-global-markets-completes-acquisition-chi-x-asia-pacific>

<sup>7</sup><https://www.cboe.com/europe/derivatives/overview/>

<sup>8</sup>[https://cdn.cboe.com/resources/release\\_notes/2021/Cboe-Europe-Derivatives-Go-Live-6th-September.pdf](https://cdn.cboe.com/resources/release_notes/2021/Cboe-Europe-Derivatives-Go-Live-6th-September.pdf) provides detailed information.

<sup>9</sup><https://www.cboe.com/europe/derivatives/european-equity-options/>

### 3 Volatility Index Interpolation

This section reviews the primary methods in use at exchanges worldwide for constructing option-implied volatility indices. They are intended to capture the expected annualized (risk-neutral) return variation from today,  $t = 0$ , until a fixed maturity date,  $t = T_M$ . However, on most days, there are no options expiring exactly in  $T_M$  calendar days, so the construction relies on volatility measures extracted from options at two neighboring maturity dates  $T_1$  and  $T_2$  with  $T_1 < T_2$ . Typically,  $T_1 \leq T_M \leq T_2$ , but this is not always feasible due to the available option maturity structure and eligibility requirements on tenors used for computing the volatility index. We measure the tenors in (calendar) days, while the volatility and variance indices, in accordance with standard practice, will be given in annualized units. Hence, the length of the measurement interval of interest,  $[0, T_M]$ , is  $\frac{T_M}{365}$  of one year.

#### 3.1 Alternative Linear Interpolation Schemes

A basic premise behind linear interpolation schemes for return variation is that the volatility measures for the available tenors are accurate. This implies that the measurement problem arises solely because, on most days, there is no measure available for the desired horizon,  $T_M$ . The existing methods then approximate the underlying “true” volatility for  $[0, T_M]$  through linear interpolation based on a pair of measures from neighboring tenors. In the case of the popular Cboe VIX index,  $T_M = 30$  calendar days or  $\frac{30}{365}$  of one year, while option tenors of less than one week are excluded, so  $T_1 \geq 7$ . Historically, the VIX was computed from monthly options for which the maturity gap is  $T_2 - T_1 = 28$  or 35 days. We denote these short and long maturity gaps by  $C_s = 28$  and  $C_\ell = 35$ , respectively. We further note that the short gap,  $C_s$ , is operative for about  $2/3$  of the maturity cycles, while the long gap,  $C_\ell$  is in effect for the remaining cycles. As a result,  $T_1$  ranges from 7 to 39 days and  $T_2$  from 35 to 67 days. In contrast, after weekly options were included in the VIX calculation during 2014, we have  $T_1 \leq 30 \leq T_2$  and  $T_2 - T_1 \leq 7$ , barring irregular market or Holiday closures.

Our analysis focuses primarily on two well-known and popular interpolation schemes, but we provide supplementary results for a third approach, common in academic studies. Of particular interest is the “Total Variance” ( $TV^2$ ) interpolation, which is, by far, the dominant technique due to the widespread adoption of the Cboe VIX-style volatility indices around the globe. An interesting alternative is the “Annualized Variance” ( $AV^2$ ) interpolation adopted by the CME Group for their recently developed suite of option-implied CVOL indices.

We denote the *annualized* option-implied return variation measures for tenors  $T_1$  and  $T_2$  by  $\sigma_1^2$  and  $\sigma_2^2$ , respectively. The objective is to generate a corresponding measure,  $\sigma_M^2$ , for the fixed target maturity  $T_M$ . As noted, in order to focus on the issues unique to maturity interpolation, we assume that the reported measures for the neighboring tenors are accurate. This implies that they represent the actual annualized total risk-neutral expected cumulative return variation over the given horizon. Letting the risk-neutral (option-implied) expectation of  $X$  formed at time  $t = 0$  by given by  $E_0^*[X]$ , we have the following relation,

$$\sigma_i^2 = \frac{365}{T_i} \cdot E_0^* \left[ \int_0^{T_i} V_u du \right], \quad \text{for } i = 1, 2, \quad (1)$$

where  $V_u$  signifies the spot return variation, reflecting the combined spot (diffusive) return variance *and* jump return variation at time  $u$ , and the factor  $\frac{365}{T_i}$  converts the option-implied cumulative variance over  $[0, T_i]$  into an annualized measure.

The annualized return variation for horizon  $T_M$  is obtained from two neighboring tenors via

linear interpolation (for  $T_1 \leq T_M \leq T_2$ ) or, if necessary (for  $T_1 > T_M$ ), extrapolation,

$$\hat{\sigma}_M^2 = w \sigma_1^2 + (1 - w) \sigma_2^2, \quad (2)$$

where the weight  $w \geq 0$ , so the measure represents a proper weighted average of the two nearby annualized measures, even if it is possible that  $w > 1$  (when  $T_1 > T_M$ ).

The  $TV^2$  and  $AV^2$  measures differ due to their distinct choice of linear weighting scheme. The former accounts for the horizon over which each measure is obtained relative to the target, thus linearly interpolating the two (annualized) cumulative (total) return variation measures, while the latter interpolates the annualized measures directly. Specifically, the two procedures adopt the following weights for equation (2), labeled respectively  $\pi$  and  $\omega$ ,

$$\pi = \frac{T_2 - T_M}{T_2 - T_1} \cdot \frac{T_1}{T_M} \quad \text{and} \quad \omega = \frac{T_2 - T_M}{T_2 - T_1}. \quad (3)$$

Thus, for fixed target maturity  $M$ , the annualized return variation measures take the form,

$$TV^2 = TV_M^2 = \pi \sigma_1^2 + (1 - \pi) \sigma_2^2 = \frac{T_2 - T_M}{T_2 - T_1} \frac{T_1}{T_M} \cdot \sigma_1^2 + \frac{T_M - T_1}{T_2 - T_1} \frac{T_2}{T_M} \cdot \sigma_2^2 \quad (4)$$

$$AV^2 = AV_M^2 = \omega \sigma_1^2 + (1 - \omega) \sigma_2^2 = \frac{T_2 - T_M}{T_2 - T_1} \cdot \sigma_1^2 + \frac{T_M - T_1}{T_2 - T_1} \cdot \sigma_2^2, \quad (5)$$

and we may omit the subscript  $M$  on  $TV$  and  $AV$ , if it equals the usual 30 day target.

If the volatility term structure is flat, i.e.,  $\sigma_1^2 = \sigma_2^2 = \sigma_M^2$ , then  $TV^2 = AV^2 = \sigma_M^2$ . Likewise, for  $T_1 = T_M$  or  $T_2 = T_M$ , we also have  $TV^2 = AV^2 = \sigma_M^2$ . Consequently, both measures work as expected in these trivial cases. Nonetheless, under empirically relevant circumstances, there is often a substantial difference between the two indices, as we demonstrate both theoretically and empirically in the following sections. For now, we simply note that the  $TV^2$  measure allocates less weight to  $\sigma_1^2$ , associated with the shorter tenor  $T_1$ , compared to  $AV^2$ .

Finally, mirroring a common approach in applied work, we introduce an approximation based on linear interpolation of the annualized standard deviations. It takes the form,

$$AS = AS_M = \omega \sigma_1 + (1 - \omega) \sigma_2. \quad (6)$$

To the best of our knowledge, this procedure is not adopted by any exchange for computation of a volatility index. Nonetheless, the simplicity and popularity of the technique renders it another natural candidate for comparison with the schemes adopted by exchanges.

### 3.2 Sources of Interpolation Bias

This section explores factors that induce discrepancies between  $TV$  and  $AV$ . For that purpose, we explore a second-order Taylor series expansion of annualized volatility around the target maturity,  $T_M$ . Of course, linear interpolation is accurate to the extent that the term structure for the return variation around the maturities  $T_1$ ,  $T_M$  and  $T_2$  is close to linear. Given the common practice in the option pricing literature of linear interpolation of implied variances and volatilities, we focus on the annualized variance measure. Hence, we expand the function

$f(T_i) = \sigma_i^2$  around  $T_i = T_M$ . Assuming  $f$  is three-times differentiable, we obtain,

$$\sigma_i^2 - \sigma_M^2 \approx f'(T_M) (T_i - T_M) + \frac{f''(T_M)}{2} (T_i - T_M)^2. \quad (7)$$

This approximation is accurate, as long as  $|\Delta T_i| = |T_i - T_M|$  is “small,” or the volatility term structure is well characterized by a quadratic function in the relevant range around  $T_M$ . On any given day, this allows for a wide range of linear, convex or concave shapes so, at a minimum, it allows us to assess the main qualitative features impacting the precision of alternative interpolation and extrapolation schemes.

Invoking the expansion (7) for both tenors  $T_1$  and  $T_2$  and ignoring third- and higher-order terms, we have the following approximation for the  $TV$  measure,

$$TV^2 - \sigma_M^2 \approx f'(T_M) [\pi \Delta T_1 + (1 - \pi) \Delta T_2] + \frac{f''(T_M)}{2} [\pi (\Delta T_1)^2 + (1 - \pi) (\Delta T_2)^2]. \quad (8)$$

The right hand side of equation (8) involves two unobserved quantities, namely the first and second derivative of the variance term structure at  $T_M$ . The remaining terms are deterministic functions of the tenors,  $T_1$ ,  $T_2$ , and  $T_M$ . Specifically, exploiting equation (3) and the definitions for  $\Delta T_1$  and  $\Delta T_2$ , direct calculations show the terms in square brackets in equation (8) equal,

$$\pi \Delta T_1 + (1 - \pi) \Delta T_2 = \frac{(T_2 - T_M)(T_M - T_1)}{T_M}.$$

and

$$\pi (\Delta T_1)^2 + (1 - \pi) (\Delta T_2)^2 = \frac{(T_2 - T_M)(T_M - T_1)(T_2 + T_1 - T_M)}{T_M}.$$

We further note that, due to the institutionally given option maturity cycle and eligibility requirements for tenors used in the index calculation, there is typically only one or two possible (predetermined) gaps between  $T_1$  and  $T_2$ ,  $C = T_2 - T_1$ . For example, as noted previously, for the VIX index, we have  $C_s = 28$  or  $C_\ell = 35$  calendar days and  $T_M$  fixed at 30 days. Hence, for the “short,” respective “long,” maturity cycle, the two quantities above are solely functions of  $T_1$ , with  $T_2 = T_1 + C_s$  and  $T_2 = T_1 + C_\ell$ . Hence, the expressions are periodic, with the identical values repeating across each short, respective long, maturity cycle.

To convey the impact of the periodically repeating values, we introduce the Convexity Adjustment Factor, CAF,

$$CAF = CAF(T_M; T_1, T_2) = \frac{(T_M - T_1)(T_2 - T_M)}{T_M^2}. \quad (9)$$

The measurement error for  $TV$  may now be written,

$$TV^2 - \sigma_M^2 \approx CAF \left[ f'(T_M) + \frac{f''(T_M)}{2} (T_2 + T_1 - T_M) \right] \cdot T_M. \quad (10)$$

To highlight the importance of each term, we introduce the following normalized quantities,

$$\tau = \frac{T_1}{T_M} \quad \text{and} \quad c = \frac{C - T_M}{T_M}.$$

Here,  $\tau$  signifies the position within the maturity cycle, and  $c$  indicates the “type” of cycle,

as determined by the relative gap between expiry dates. Exploiting this notation, we obtain,

$$CAF = (1 - \tau)(\tau + c), \quad (11)$$

and the approximate error of the  $TV$  measure in equation (10) may be restated as,

$$TV^2 - \sigma_M^2 \approx (1 - \tau)(\tau + c) [f'(T_M) + f''(T_M) (\tau + \frac{c}{2}) T_M] T_M. \quad (12)$$

First, we note that  $c$  typically will be small. For example, for the traditional VIX measure based on monthly options,  $c_s = (28 - 30)/30 = -1/15$  and  $c_\ell = (35 - 30)/30 = 1/6$ , so  $\tau + c$  is positive for all relevant  $\tau$ . In fact, to a decent approximation, one may set  $c = 0$ . Nonetheless, this term is key to account for systematic differences in the measurement errors for short and long maturity cycles. Second, it is evident that the error tends to shrink as either of the two maturities approaches  $T_M$ , i.e., when  $\tau \approx 0$  or  $\tau \approx 1$ , as this reduces the impact from CAF. Third, we see that the two terms in the square brackets will offset each other, when  $f'(T_M)$  and  $f''(T_M)$  are of opposite sign, but will amplify each other, if they are of the same sign. The latter observation is particularly relevant, as we often find the variance term structure to be moderately upward sloping, but concave, during normal times, and then strongly downward sloping and convex during crisis or turbulent market conditions. Finally, we note that the measurement error, all else equal, changes the sign as  $\tau$  moves across unity and that this error is augmented through the second derivative, when  $\tau$  is large. In Section 4, we confront these predictions with actual patterns from data on MFIV measures.

Next, we generate the corresponding results for the  $AV$  measure in equation (5). Since this procedure involves direct linear interpolation, it will by construction not feature any first-order Taylor series bias. It is straightforward to derive the following local approximation,

$$AV^2 - \sigma_M^2 \approx (1 - \tau)(\tau + c) [f''(T_M) / 2] T_M^2. \quad (13)$$

Hence, not surprisingly, the  $AV$  measurement error primarily reflects the distance of the two tenors from  $T_M$  and the curvature in the variance term structure. Likewise, as for  $TV$ , the sign of the bias will shift when  $\tau$  moves across unity, and the bias is especially large, when  $\tau$  is substantially above one. Finally, the bias also grows with  $c$ , the relative maturity gap.

### 3.3 The VIX Convexity Adjustment Factor

It is evident from equations (12) and (13) that the periodic pattern in  $TV$  and  $AV$  is induced by CAF, subject only to the natural assumption that the variance term structure is unrelated, or exogenous, to the institutionally determined option tenors.<sup>10</sup> Figure 3 depicts the CAF values for VIXMO and VIXWE across our sample. The pronounced deterministic cyclical pattern in VIXMO is evident, while the VIXWE values are negligible in comparison. The more extreme outliers generated for the long versus short VIXMO maturity cycles are also readily observed, reflecting the difference generated by  $c = c_s$  versus  $c = c_\ell$ . The display renders the origin of the October 2014 structural break transparent. The VIX computation relies on the highly volatile (red) VIXMO CAF values up to that point, and the (blue) stable VIXWE values thereafter.

Even if CAF varies substantially within each VIXMO cycle, it does so quite smoothly. For the short (long) cycle, the maximum of 0.22 (0.34) is obtained for  $T_1 = 16(13)$  days, and the

<sup>10</sup>The maturity structure for options is established by the exchange and altered infrequently, typically remaining invariant for years. Hence, the tenor structure is predetermined relative to the ongoing option valuation.

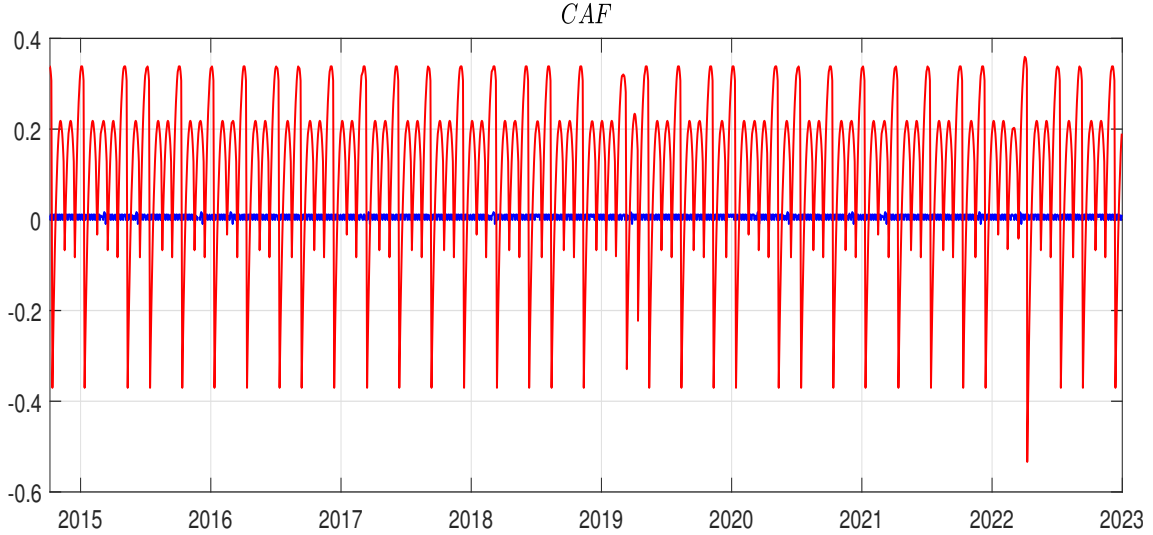


Figure 3: Convexity Adjustment Factor (CAF) for VIXMO (red) and VIXWE (blue).

minimum for the largest  $T_1$ , i.e., 32 (39) days, with a CAF of -0.07 (-0.37). Thus, the periodic CAF pattern will not cause an abrupt VIX shift within the cycle, even if it may induce a bias.

At roll-over, however, one cycle ends and a new one is entered. Then CAF jumps from its value at  $\tau = 7/30$  to the value for  $\tau = 32/30$  (39/30), if the preceding cycle was short (long). Specifically, in the short-to-short (VIXMO) transition, CAF changes from 0.13 to -0.07, for a jump of -0.20. The corresponding change for the long-to-short roll-over is 0.31 to -0.37, for a jump of -0.68. Finally, the short-to-long transition induces a CAF jump from 0.13 to -0.08, or -0.21, which is similar to the effect for the short-to-short roll-over. We later explore the significance of these abrupt shifts in CAF for the reported VIX(MO) values.

Importantly, as noted, Figure 3 also shows that – due to the proximity of the tenors for the VIXWE calculation to the target maturity – the corresponding CAF values are minuscule. Hence, it is practically free of interpolation error. In turn, this implies that VIXWE provides an accurate proxy for the VIXMO values that would be observed if interpolation errors were absent. Consequently, we have an operational way to circumvent the obstacles posed by the unknown values for the slope and curvature of the volatility term structure, appearing in equations (12) and (13). Whenever it is feasible to compute a VIXWE measure we can, with a high degree of precision, approximate the interpolation error for VIXMO via its discrepancy from the concurrent VIXWE value. No such “model-free” assessment of the interpolation error is feasible without the presence of liquid weekly option markets, and the associated VIXWE measure.

### 3.4 Interpolation Bias in the Heston Model

As a benchmark for our analysis, we explore the interpolation bias for VIXMO arising within an empirically reasonable, yet tractable, representation of the risk-neutral volatility dynamics, the one-factor Heston stochastic Volatility (SV) model. This setting allows for a closed-form solution for the bias, providing a guide to assess potential sources of measurement error.

This SV model stipulates that the instantaneous variance follows a mean-reverting affine representation under the risk-neutral measure,

$$dV_t = \kappa(\theta - V_t) dt + \nu \sqrt{V_t} dW_t^Q. \quad (14)$$

Given this parametric stochastic volatility specification, we have,

$$\sigma_t^2 = E_0^{\mathbb{Q}} \left[ \frac{1}{t} \int_0^t V_u ds \right] = \theta + (V_0 - \theta) \cdot \frac{1 - e^{-\kappa t}}{\kappa t}.$$

For a fixed  $\kappa > 0$ , it is convenient to define an auxiliary function,

$$g(t) = g(t; \kappa) = \frac{1 - e^{-\kappa t}}{\kappa t} = 1 - \frac{\kappa t}{2} + O(\kappa^2 t^2).$$

We note that for  $t > 0$  the function  $g(t)$  is decreasing and convex,  $g'(t) < 0$  and  $g''(t) > 0$ . The approximation errors vis-a-vis the true variance,  $\sigma_M^2$ , equal

$$TV^2 - \sigma_M^2 = (V_0 - \theta) [\pi g(T_1) + (1 - \pi) g(T_2) - g(T_M)], \quad (15)$$

$$AV^2 - \sigma_M^2 = (V_0 - \theta) [\omega g(T_1) + (1 - \omega) g(T_2) - g(T_M)]. \quad (16)$$

**Proposition 1** *In the one-factor Heston SV model, the measurement errors in equations (15) and (16) always have opposite signs.*

*Specifically, for an **upward** sloping term structure of volatility ( $V_0 < \theta$ ),*

$$(1) \text{ when } 0 < \pi, \omega < 1 \text{ (interpolation): } TV^2 - \sigma_M^2 > 0 \text{ and } AV^2 - \sigma_M^2 < 0,$$

$$(2) \text{ when } \pi, \omega > 1 \text{ (extrapolation): } TV^2 - \sigma_M^2 < 0 \text{ and } AV^2 - \sigma_M^2 > 0.$$

*In the case of a **downward** sloping volatility term structure ( $V_0 > \theta$ ), the signs are reversed in all of the above inequalities.*

*Proof:* The results for  $AV^2 - \sigma_M^2$  follow from the convexity of the function  $g(t)$ . To see this, note that  $\omega T_1 + (1 - \omega) T_2 = T_M$ , so equation (16) and the convexity of  $g(t)$  directly imply the posited inequalities for  $AV$ . A similar argument works for  $TV$ , but now because of the concavity of the function  $h(t) = t g(t)$  for  $t > 0$ . From the definition of the weight  $\pi$ , we have,

$$\begin{aligned} \pi g(T_1) + (1 - \pi) g(T_2) - g(T_M) &= \omega \frac{T_1}{T_M} g(T_1) + (1 - \omega) \frac{T_2}{T_M} g(T_2) - g(T_M) \\ &= \frac{1}{T_M} [\omega h(T_1) + (1 - \omega) h(T_2) - h(T_M)]. \end{aligned}$$

□

Figure 4 illustrates how the results in Proposition 1 manifest themselves in a couple of typical scenarios. The left panel of the top row displays two different volatility term structures - one downward and one upward sloping - associated with a high and low initial volatility state of 40% and 10%, respectively, when the unconditional mean volatility is 20%. The VIXMO CAF values are depicted as a function of  $T_1$  in the right panel of the top row. Depending on the point in the maturity cycle, these CAF values induce a certain interpolation bias for the interpolated 30-day VIX values, as shown in the bottom row. The left panel refers to the upward sloping term structure with a model-determined 30-day VIX of 12.44%. Here, the  $TV$  approach generates a sizable positive bias, whenever  $T_1$  is below  $T_M = 30$  days, while the negative bias for  $T_1$  above

30 days turns sharply negative, as  $T_1$  grows. In comparison, the  $AV$  biases are much smaller and, as stated in results (1) and (2) of Proposition 1, they are of the opposite sign of the  $TV$  bias. Finally, for completeness, we also display the bias from the  $AS$  interpolation technique. It mimics the behavior of  $AV$  quite closely, although the bias, uniformly, is slightly larger. The right panel depicts the downward sloping term structure scenario with a model-determined 30-day VIX value of 37.17%. Here, the mirror image appears, with all biases changing sign, consistent with the result in the proposition regarding a downward sloping term structure. The relative biases remain largely unaltered, except for the  $AS$  bias now being, uniformly, slightly smaller than for  $AV$ . Hence,  $TV$  looks unattractive in both cases, generating a substantially larger bias than the other commonly used approaches.

Of course, the biases in Figure 4 reflect the underlying choice of parameter values, with the strength of mean reversion  $\kappa$  being the key coefficient governing the nonlinearity of the term structure. To provide a broader sense of the interpolation errors for  $TV$  versus  $AV$  in this setting, Figure 5 depicts the relative error for four distinct values of  $\kappa$ , where large values imply stronger mean reversion along with a steeper slope and a more pronounced curvature in the term structure. The relative error, defined below, takes negative values, when the interpolation errors for  $TV$  exceed those for  $AV$ ,

$$RE = \frac{\Delta_{AV}}{\Delta_{TV}} - 1.$$

Figure 5 reveals that  $AV$  is universally preferable compared to  $TV$  for the one-factor Heston model. Moreover, as the slope and curvature increases with  $\kappa$ , the errors compound for  $TV$ , rendering the traditional VIXMO procedure increasingly uncompetitive. These findings serve as a benchmark for our empirical findings in Section 4, even if there are important scenarios for which these conclusions do not apply.

## 4 Empirical Assessment of the VIX Interpolation Errors

In this section, we first briefly describe the data used in our empirical analysis. We then construct proxies for the VIXMO interpolation errors exploiting the fact, established in Section 3.3, that the corresponding VIXWE errors, in comparison, are negligible. This sheds light on the extent of the structural change in the VIX following the conversion from VIXMO to VIXWE in October 2014. Moreover, as documented in Section 2.2, the vast majority of existing MFIV measures adheres to the VIXMO interpolation approach. Hence, our findings offer insights relevant across the full universe of VIX-type indices. Specifically, they address the impact of the option maturity structure, the trade-offs associated with the choice of roll-over date from nearby to longer tenors in the index computation, and the use of  $TV$  versus  $AV$  weights for maturity interpolation.

### 4.1 The SPX Option Sample

We observe all daily closing quotes for the full panel of SPX options over the sample period 2004-2023. Early on, only monthly and quarterly options were traded, typically generating just one single monthly expiration, scheduled for the third Friday of the month. Over time, however, the set of SPX options expanded, and the listing of six or more consecutive end-of-week Friday expiration options has been in effect since June 2012. This ensures the presence of options with expiries straddling the target 30 calendar days, and with only seven days separating the tenors, at all times, barring Holidays. Moreover, longer-dated options continue to be offered. Consequently, for every trading day since March 2012, we are able to compute the standard



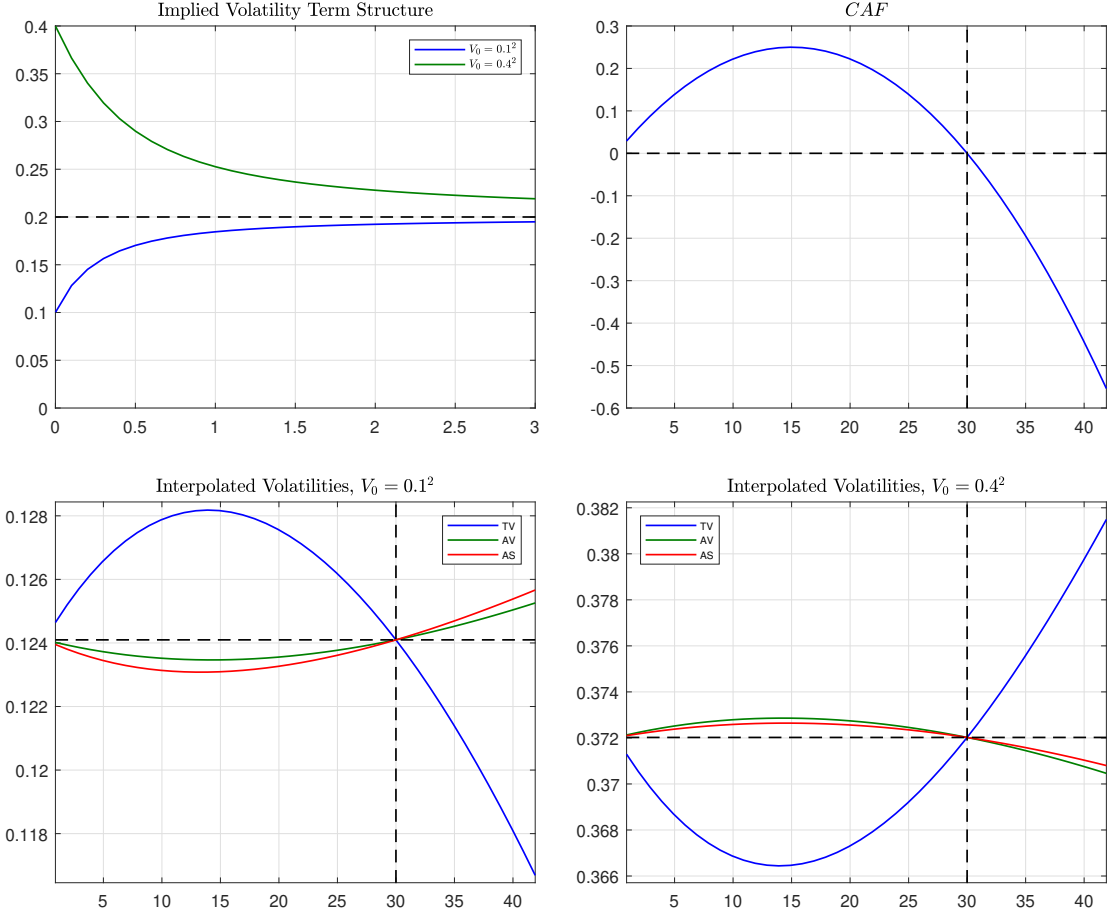


Figure 4: The illustration uses model (14) with  $\kappa = 5$ ,  $\theta = 0.2^2$ , and an initial low or high level of spot volatility,  $V_0 = 0.10^2$  and  $V_0 = 0.40^2$ . The top left panel displays the two resulting volatility term structures in the Heston model. The remaining panels plot various measure against the first maturity  $T_1$ , assuming  $T_2$  is 30 calendar days beyond  $T_1$ . The vertical dashed line separates the extrapolation ( $\omega, \pi < 0$ ) and interpolation ( $\omega, \pi > 0$ ) scenarios. The top right panel plots the CAF, while the bottom panels display the interpolated volatilities  $TV$ ,  $AV$  and  $AS$  for the downward and upward sloping term structures, respectively, while the horizontal dashed black line indicates the true 30-day volatility,  $\sigma_M$ .

VIX measure based on monthly options along with a corresponding VIX measure from weekly options, i.e., we obtain both a VIXMO and a VIXWE measure each trading day.

In fact, going one step further, the presence of weekly options covering a period beyond six weeks allows us to provide a good approximation to the full MFIV term structure for the range of 2-70 calendar days. This opens up for additional opportunities to assess alternative interpolation techniques for volatility indices based on different option tenors.

## 4.2 Illustrative Interpolation Examples

This section exemplifies how major VIX interpolation errors materialize. For a pair of specific trading days, we assess the plausible range for the “correct” VIX index value, given the corresponding observations at neighboring maturities, using the predictions from the one-factor Heston model to guide our interpretation.

For our illustration, we turn to a turbulent period, namely Friday and Monday, October 10 and 13, 2008. The first row of Figure 6 captures the VIX computational procedure at the

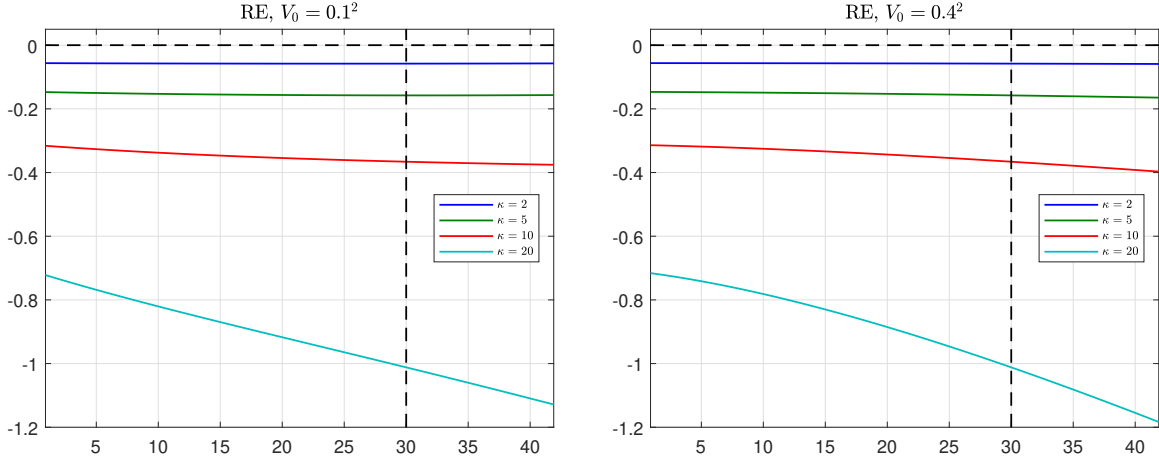


Figure 5: This figure plots the relative error  $RE$ . All parameters are as in Figure 4, except for  $\kappa$ , which takes one of the four values (2, 5, 10, and 20).  $RE$  for  $T_1 = 30$  is determined by l'Hospital's rule.

close on Friday, while the second row depicts the interpolation on the following trading day (Monday). The strongly downward sloping volatility term structure and the particular option maturity structure on that date render the distortions especially transparent. October 10 was the last trading day prior to roll-over, with option tenors of  $T_1 = 7$  days and  $T_2 = 42$  days. On the following Monday, the VIX was instead computed via extrapolation, using options with tenors of  $T_1 = 39$  and  $T_2 = 67$  days.

The left panels of Figure 6 display the implied return variation in annualized volatility terms. In the top row, the VIX is determined by Cboe via interpolation and, in line with the prediction in Proposition 1,  $TV$  generates a lower measure than  $AV$ . The linear (annualized) “sigma” interpolation scheme  $AS$  is also depicted and, for the relevant range around 30 days, it is near identical to the  $AV$  measure (which performs linear interpolation in annualized variance terms). In the corresponding right panel, the return variation is given in total variance terms, and now the  $TV$  procedure is obtained via linear interpolation, while the  $AV$  and  $AS$  procedures are highly nonlinear in this metric.

Evidently, the choice of interpolation scheme has a material impact on the MFIV measure. On Friday,  $TV$  produces a VIX value of 72.2% versus an  $AV$  value of 81.3%. The ranking flips on Monday, when  $TV$  and  $AV$  generate values of 54.9% and 53.1%, respectively, again consistent with Proposition 1 for the case of a downward sloping term structure and VIX extrapolation. For each day, it is natural to take the two values as extreme points and deem the correct VIX value to fall between the  $TV$  and  $AV$  measures. However, assuming the Heston model provides useful guidance, we obtain an additional set of bounds on the VIX value. For example, on October 10, if we compute MFIV from the  $T_2 = 42$  and  $T_3 = 70$  day tenors via extrapolation, we obtain  $TV$  and  $AV$  measures of 77.0% and 72.7%, respectively. Proposition 1 now stipulates that the VIX lies between these two values, representing a substantive narrowing for the range of potential VIX values to [72.7%, 77.0%]. Figure 7 depicts the  $T_1$  and  $T_2$  interpolation versus the  $T_2$  and  $T_3$  extrapolation on October 10 in the left and right panels, respectively. A similar comparison for October 13 shows that [53.1%, 54.9%] is the most accurate assessment feasible using our simple guide from the one-factor Heston model.

This illustration shows that the October 10 VIX ( $TV$ ) value of 72.2% falls below the range obtained using interpolation from different tenor constellations and the assumptions in Proposition 1. However, this does not fully corroborate the tentative conclusion in Section 3.4, that  $TV$

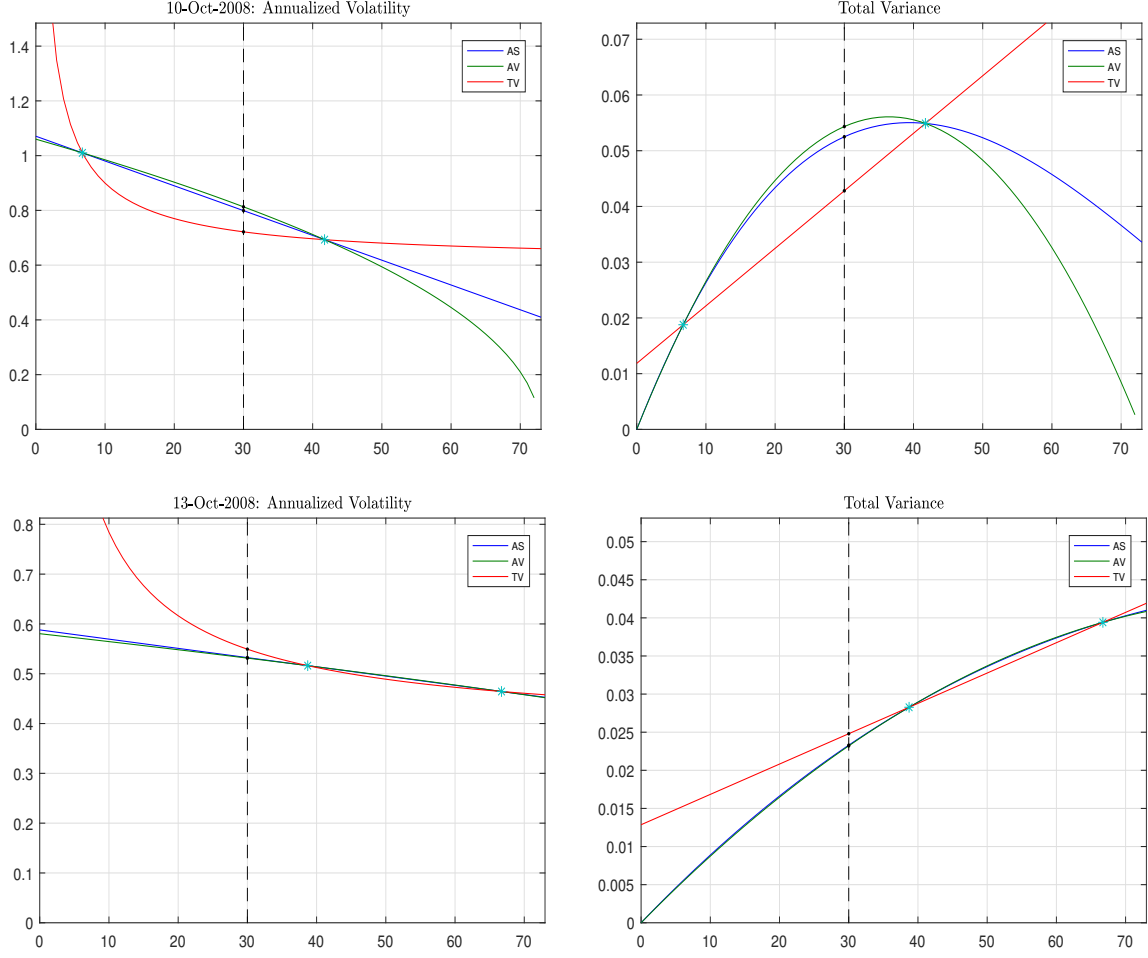


Figure 6: *AV* and *TV* interpolation on two selective days. For the first day,  $T_1=7$  and  $T_2=42$  days. For the second day (which is the next trading day),  $T_1=39$  and  $T_2=67$  days.

is inferior to alternative interpolation techniques. Specifically, the *AV* interpolation on October 10, using the prescribed tenors of 7 and 42 days, generates a dramatic inflation in the MFIV value, reaching 81.3% which, compared to *TV*, is starkly at odds with the bounds established via Proposition 1, so *AV* looks particularly suspect in this scenario. It also suggests that the one-factor Heston model fails to provide a good approximation to the variance term structure in this case. Finally, for each panel in Figure 7, the three MFIV values for the available SPX option expiries are indicated by green dots. Clearly, interpolation techniques inducing a perfect MFIV fit for two of these maturities will miss the third MFIV value by a large margin, highlighting the potential failures of all standard linear interpolation techniques.

These illustrations suggest that the *TV*-based VIX measure on occasion is subject to substantial distortions in the level and, especially, in the daily VIX innovation around the roll-over date. For October 10-13, 2008, our analysis points towards a downward bias in MFIV on Friday and upward bias on Monday, suggesting that the drop in implied volatility over the three-day period is severely understated at the officially disseminated VIX values.

Motivated by this conclusion, we now turn to a direct empirical investigation of the relative performance of the alternative interpolation techniques, exploiting the precision offered by the availability of weekly options from 2012 onwards.

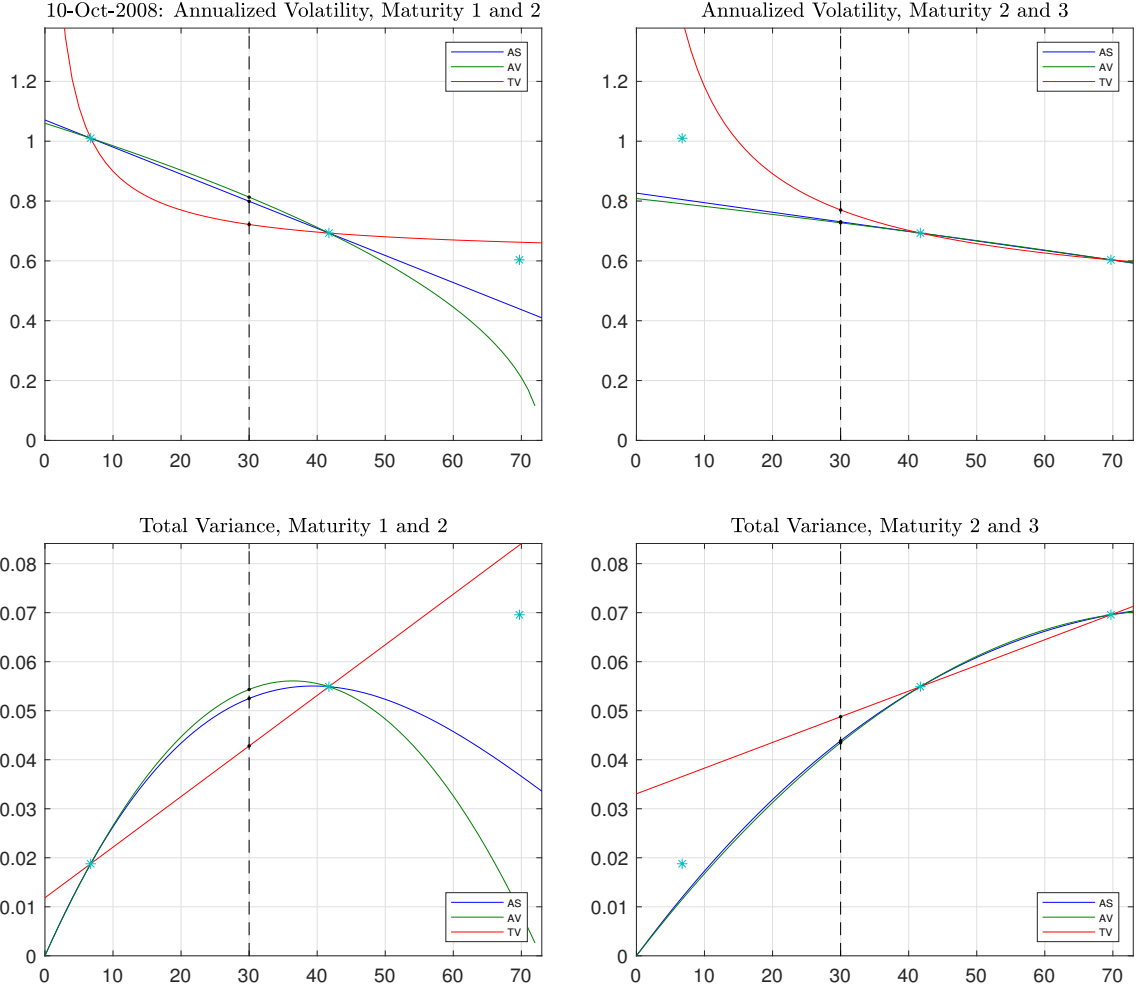


Figure 7:  $AV$  and  $TV$  interpolation on one selective day, for which  $T_1=7$ ,  $T_2=42$ , and  $T_3=70$  days.

### 4.3 Assessing the Realized VIX Interpolation Errors

We first seek to quantify the typical error in VIXMO stemming from the interpolation technique. Our procedure is straightforward. We have observations for both monthly and weekly options from March 2012 till December 2020. For every trading day in this subsample, we compute  $TV$ ,  $AV$ , and  $AS$  based 30-day MFIV measures using the tenors underlying VIXWE and VIXMO. The  $TV$  measures will be denoted  $TV_{(W)}$  and  $TV_{(M)}$ , with a similar notation adopted for the corresponding  $AV$  and  $AS$  interpolated measures. Because the VIX is  $TV$ -based, we have that  $TV_{(M)} \approx \text{VIXMO}$  and  $TV_{(W)} \approx \text{VIXWE}$ .<sup>11</sup> Moreover, because the VIXWE errors are negligible, we simply treat the VIXMO - VIXWE discrepancy as the VIXMO interpolation error.

Given that the potential errors are related to the option maturity structure through its manifestation in the CAF measure, Figure 8 displays the average interpolation error, proxied by the excess of the  $TV_{(M)}$  and  $AV_{(M)}$  measures over VIXWE, along with the corresponding mean absolute deviation (MAD) as a function of the nearby option tenor  $T_1$ .

Some systematic effects are readily identified, despite the noise stemming from having at

<sup>11</sup>For an analysis of market frictions inducing deviations between the official VIX index and the corresponding empirical  $TV$  measure computed from SPX option quotes, see Andersen, Bondarenko & Gonzalez-Perez (2015).

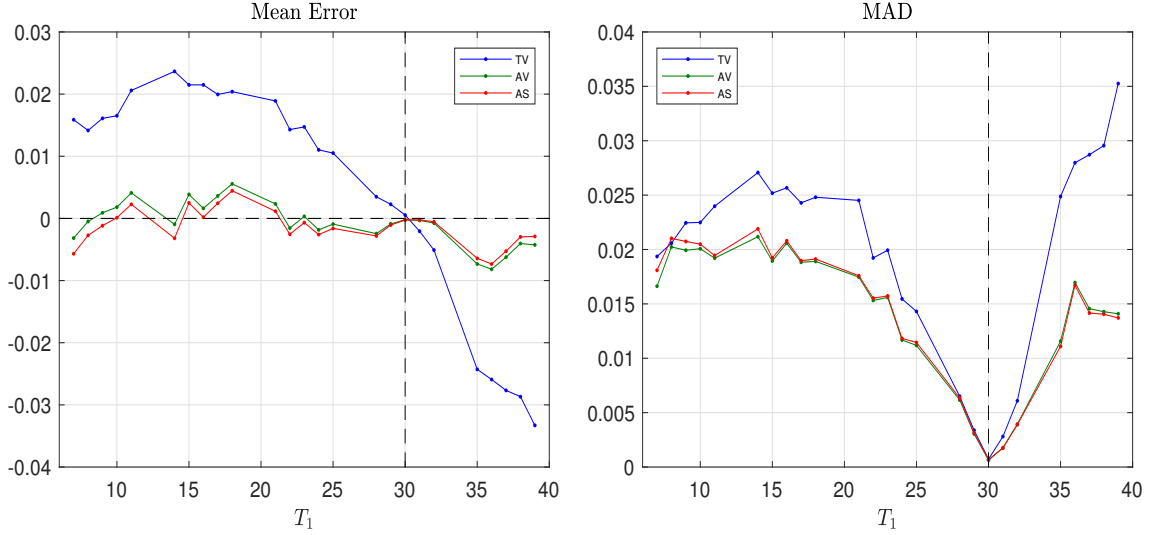


Figure 8: **Left** panel: Mean interpolation errors of alternative VIXMO proxies relative to VIXWE as a function of the nearby maturity  $T_1$ . **Right** panel: The corresponding mean absolute deviation. The sample covers May 2012-Dec 2020.

most one observation available for each  $T_1$  value per maturity cycle. On average, or unconditionally, the *AV* and *AS* measures are essentially unbiased, while the *TV* interpolation induces a substantial positive bias for low  $T_1$  values and negative biases when VIX is obtained via extrapolation, i.e.,  $T_1 > 30$  days. These observations are in line with the conjecture in Section 3.4 that *TV* may underperform relative to *AV* and *AS*. However, the MAD measures for *TV* and *AV* are quite closely aligned, indicating that there also is a substantial degree of variation in the *AV* measure around the VIXWE value. One striking illustration of a large *AV* interpolation error is the October 10, 2008, episode discussed in Section 4.2.

These impressions are corroborated by Table 3, which provides summary error statistics across five tenor categories. Only for the extrapolation group do we see a dramatic gap between the MAD of *TV* and *AV*. In other words, even if *AV*-interpolation provides unconditionally unbiased VIX measures, there is a large degree of error variation, stemming from significant episodic errors. Finally, the table confirms that, in practice, *AV* and *AS* are effectively identical.

| $T_1$   | $N$ | <i>TV</i> |       |       | <i>AV</i> |       |       | <i>AS</i> |       |       |
|---------|-----|-----------|-------|-------|-----------|-------|-------|-----------|-------|-------|
|         |     | Mean      | StDev | MAD   | Mean      | StDev | MAD   | Mean      | StDev | MAD   |
| Group 1 | 492 | 0.017     | 0.023 | 0.022 | 0.001     | 0.026 | 0.019 | -0.001    | 0.027 | 0.020 |
| Group 2 | 468 | 0.021     | 0.024 | 0.025 | 0.003     | 0.025 | 0.020 | 0.001     | 0.026 | 0.020 |
| Group 3 | 473 | 0.014     | 0.020 | 0.019 | -0.000    | 0.020 | 0.014 | -0.001    | 0.020 | 0.014 |
| Group 4 | 484 | -0.000    | 0.006 | 0.004 | -0.001    | 0.005 | 0.003 | -0.001    | 0.005 | 0.003 |
| Group 5 | 182 | -0.028    | 0.023 | 0.029 | -0.006    | 0.018 | 0.014 | -0.005    | 0.018 | 0.014 |

Table 3: **Interpolation Errors.** Relative errors,  $X/Y - 1$ , where  $X$  is a VIXMO proxy obtained by interpolation via  $TV_{(M)}$ ,  $AV_{(M)}$ , or  $AS_{(M)}$ , while  $Y = \text{VIXWE}$ . Statistics are computed for tenor Group 1-5, defined by  $T_1$  belonging to the intervals  $[7, 11]$ ;  $[14, 18]$ ;  $[21, 25]$ ;  $[28, 32]$ ; and  $[35, 39]$ , respectively. MAD is computed as the absolute deviation from VIXWE. The sample covers May 2012-December 2020.

The illustration in Section 3.4 centers on a roll-over date. In fact, equations (12)–(13) imply that the largest errors occur when CAF is elevated and the variance term structure is steep and heavily curved (nonlinear). This motivates our subsequent analysis of interpolation errors around roll-overs. An important limitation is the sparsity of such events, as we have at most one observation per month, and they are further spread across transitions between different option maturity cycles, generating distinct values of  $c$  in equations (12)–(13). More robust measures of roll-over distortions would be available, if we have multiple observations of differently shaped term structures, each coinciding with the roll-over. However, the term structure typically displays strong nonlinearity only during extreme market conditions and such events are, almost by definition, rare, so we face a small-sample problem. In the next section, we develop an approach to mitigate this concrete problem.

#### 4.4 Improved Measures of Interpolation Errors via Historical Simulation

The introduction of weekly options, detailed in Table 1, enables us to construct a good approximation to the risk-neutral volatility term structure for tenors spanning the full 2-70 calendar day range. Under the natural assumption that the option maturity structure is independent of the true volatility process, we may then assign a MFIV value to each such tenor every trading day. Armed with these values, we may compute  $TV_{(M)}$ ,  $AV_{(M)}$ , and  $AS_{(M)}$  proxies for VIXMO and VIXWE. That is, every trading day provides “pseudo” observations across all relevant  $(T_1, T_2)$  combinations of interest. Thus, since 2012, we may generate a set of daily interpolation errors for all relevant tenors and maturity cycles. We label this approach *historical simulation*, as it relies on MFIV measures extracted from actual weekly option quotes.

To facilitate the subsequent exposition concerning interpolation biases around roll-over events, we introduce some additional notation. We let  $\mathbf{T} = (T_a, T_b)$ ,  $0 < T_a < T_b$ , denote a generic pair of tenors used to approximate the MFIV at some target maturity date,  $T_M$ . We also allow for the choice of tenors to be explicit in the notation for the interpolation scheme, e.g., for  $\mathbf{T} = (T_1, T_2)$  referring to the two maturities used to compute the TV proxy for VIXMO, we may write  $TV_{(M)}(\mathbf{T}) = TV_{(M)}(T_1, T_2)$ . In particular, we shall make use of the following shorthand for the tenors used around a roll-over event.  $\mathbf{T}_{E,S} = (7, 35)$  will - barring Holidays on the respective expiries - denote the tenors for the roll-over date at the end of a short monthly option maturity cycle, while  $\mathbf{T}_{E,L} = (7, 42)$  refers to tenors at the roll-over date of a long cycle. Likewise, as the nearby short or long monthly cycle reaches the roll-over date, the tenors for the subsequent monthly cycle are  $\mathbf{T}_{B,S} = (35, T_3)$  and  $\mathbf{T}_{B,L} = (42, T_3)$ , where  $T_3$  denotes the third smallest monthly option maturity date.<sup>12</sup> Finally, the tenors for the first trading day following the roll-over will typically be  $\mathbf{T}_{R,S} = (32, T_3 - 3)$  or  $\mathbf{T}_{R,L} = (39, T_3 - 3)$ , depending on whether the preceding cycle was short or long, and the subtraction by three calendar days reflects the gap between the Friday prior to roll-over and the following Monday.

As indicated in Section 4.3, interpolation errors are most likely to induce abrupt shifts in the VIX series around roll-over event. To investigate this hypothesis, we compute a proxy for the VIX roll-over “gross return,” and decompose it into two conceptually distinct components. For illustration, assume we apply the TV interpolation technique and the roll-over occurs at the end of a short monthly cycle between a Friday close and a Monday open. Analogous definitions apply for alternative interpolation techniques and for roll-overs following long monthly cycles.

<sup>12</sup>If the new monthly option cycle is short, we have  $\mathbf{T}_{B,S} = (35, 63)$  and if it is long, we have  $\mathbf{T}_{B,S} = (35, 70)$ . Since a long cycle is almost always followed by a short cycle, we typically have  $\mathbf{T}_{B,L} = (42, 70)$ . The interpolation errors are not very sensitive to these distinct scenarios, so we avoid introducing additional notation for the tenor combinations across these distinct cases.

$$\begin{aligned}
R_{TV}^{(S)} &= \frac{TV_{(M)}(\mathbf{T}_{R,S})}{TV_{(M)}(\mathbf{T}_{E,S})} = \frac{TV_{(M)}(32, T_3 - 3)}{TV_{(M)}(7, 35)} \\
&= \frac{TV_{(M)}(\mathbf{T}_{B,S})}{TV_{(M)}(\mathbf{T}_{E,S})} \times \frac{TV_{(M)}(\mathbf{T}_{R,S})}{TV_{(M)}(\mathbf{T}_{B,S})} \\
&= \text{Pure Roll-Over Effect} \times \text{Shrinking Tenor Effect} .
\end{aligned}$$

To be even more specific, assume the short cycle rolls into a long cycle. In the above example, the “pure roll-over” effect then captures the contemporaneous impact of switching from the first short to the second long monthly cycle, i.e.,  $TV_{(M)}(35, 70)/TV_{(M)}(7, 35)$ , while the “shrinking tenor” effect signifies the impact of the declining tenor between Friday and Monday, given by  $TV_{(M)}(32, 67)/TV_{(M)}(35, 70)$ . Notice that the former effect reflects only the shift in tenor, while the latter is subject to intermediary and potentially large idiosyncratic movements in volatility between Friday and Monday, and thus subject to a larger degree of random sample variation.

Exploiting the historical simulation approach, we now generate a large sample of “synthetic” roll-over events, namely one every trading day since May 2012. Of course, the results are impacted by the specific variation observed in the volatility term structure over this sample. However, the procedure ensures that all interesting configurations materializing across the period are reflected in our estimates for all tenor combinations of interest.

Table 4 reports the average total roll-over return, its decomposition, and associated standard deviations for  $TV$  and  $AV$  interpolation based on our historical simulation approach. Since there is non-trivial heterogeneity depending on the maturity cycle, the table separates rolling out of a short and a long cycle. Specifically, the -4.97% entry for **TV-L** in the top row refers to the mean total TV roll-over return,  $R_{TV}^{(L)} - 1$ . Since the historical simulation approach generates one (pseudo) roll-over return each trading day, the effect is measured accurately, verifying that these specific transitions induce a sizable mean drop in our VIX proxy which, in reality, occur about four times annually. The corresponding mean return for short cycle roll-over,  $R_{TV}^{(S)} - 1$ , is a more moderate, yet non-trivial, -1.23%, with these events taking place roughly eight times a year. In comparison, the average  $AV$  total roll-over returns are all close to zero.

Figure 8 captures the qualitative forces at work. As we jump from  $T_1 = 7$  to  $T_1 = 32$  in rolling out of a short maturity cycle, we move from an average large positive interpolation error to a small negative one, inducing the -1.23% average percentage move. However, the isolated effect of shifting to the next cycle is even larger. If we had rolled prior to the nearby maturity reaching  $T_1 = 7$ , then the interpolation would exploit  $T_1 = 35$  along with a  $T_2$  tenor of 63 (70), when the following maturity cycle is short (long). Thus, the immediate impact of rolling at the end of the trading day can be assessed using the change from  $T_1 = 7$  to  $T_1 = 35$ . The mean of this pure roll-over “return” is reported in the second row of Table 4 and amounts to -2.34%. The gap between the first and second row reflects the drop in tenor between Friday and Monday. Since the negative TV interpolation error is rapidly declining over that range, this induces the mean percentage gain of 1.16%, reported in the third row. The analogous decomposition of the roll-over effect is provided for the alternative roll-over scenarios and interpolation techniques in the table as well. Most strikingly, the “pure” roll-over effect for TV around the conclusion of a long cycle is -6.45%. We further note that the two components of the  $AV$  roll-over returns display minuscule average changes.

The small mean percentage distortions induced by  $AV$  interpolation support the conjecture that  $AV$ , on average, is superior to  $TV$ , also around roll-over events. However, the large standard

|                 | TV – S |       | AV – S |       | TV – L |       | AV – L |       |
|-----------------|--------|-------|--------|-------|--------|-------|--------|-------|
|                 | Mean   | StDev | Mean   | StDev | Mean   | StDev | Mean   | StDev |
| Total Roll-over | -1.23  | 8.35  | 0.09   | 8.51  | -4.97  | 9.04  | 0.17   | 8.88  |
| Pure Roll-over  | -2.34  | 1.65  | -0.17  | 1.77  | -6.45  | 4.33  | -0.04  | 3.44  |
| Shrinking Tenor | 1.16   | 8.60  | 0.25   | 8.29  | 1.67   | 9.30  | 0.19   | 8.07  |

Table 4: **VIXMO jumps induced by Maturity Roll-Over.** The top row of the table reports mean roll-over percentage returns for  $TV_{(M)}$  and  $AV_{(M)}$  interpolation, separating the cases where we roll out of a short (S) and long (L) cycle, with **S** - Short Cycle ( $T_2 - T_1 = 28$ ) and **L** - Long Cycle ( $T_2 - T_1 = 35$ ). The returns associated with the pure roll-over and shrinking tenor effects are provided in the second and third row. Historical simulation is used to estimate the returns over May 2012 - Dec 2020.

deviations imply that the  $AV$  measures also will tend to generate large errors. Furthermore, they reveal that the main source of fluctuations in VIX measures around the roll-over is the tenor effect. The pure roll-over effect, reported in the second row, strips out the impact of economic events on volatility across the three day roll-over window, thus providing a cleaner measure of the roll-over effect *per se*, and identifies the large impact associated with roll-overs from long cycles. Without the ability to exploit the historical simulation approach and the decomposition, this effect would be near impossible to determine, given the rare occurrences of long cycle roll-overs.

In summary, Table 4 confirms that TV-style VIXMO measures are subject to substantial distortion, generating an average 6.45% negative impact on the volatility measure for long cycle roll-overs, and a corresponding 2.34% negative error, when rolling out of a short cycle. These effects are partially masked by a mean positive bias in excess of one percent across the three-day roll-over window. In contrast, the  $AV$  measure is largely free of such mean distortions. Nonetheless, the  $AV$  procedures does induce spurious variation in the volatility measure. For the long cycle roll-over events, the standard deviation of 3.44% testifies to a substantial variation attributable to the shifting shape of the volatility term structure. This confirms again that the  $AV$  approach also is particularly error-prone around the roll-over date.

To further explore whether the roll-over events are particularly problematic, the left panel of Figure 9 displays the average interpolation biases relative to VIXWE on roll-over dates as well as regular (non roll-over) trading days, using the historical simulation technique. For all interpolation procedures, regular days do not display any average bias. For the  $TV$  measures, this may be surprising, but reflects a cancellation of the positive errors for  $T_1 < 30$  calendar and negative errors for  $T_1 > 30$ . However, the  $TV$  interpolation biases on short (Group 2), and especially, long (Group 3) roll-over dates are striking. The right panel of the figure also reveals the elevated degree of fluctuation in the errors for roll-over dates.

#### 4.5 Interpolation Errors Induced by the Volatility Term Structure

We have explored the systematic interpolation errors as a function of the option tenors, as succinctly summarized by the CAF measure. Furthermore, our roll-over analysis in Section 4.4 documents that these errors, empirically, are more pronounced when the volatility term structure is highly nonlinear. This section investigates in more depth how the shape of the term structure of volatility impacts the interpolation errors. We again rely on historical simulation to obtain an adequate sample size, using data from 2012 and forward.



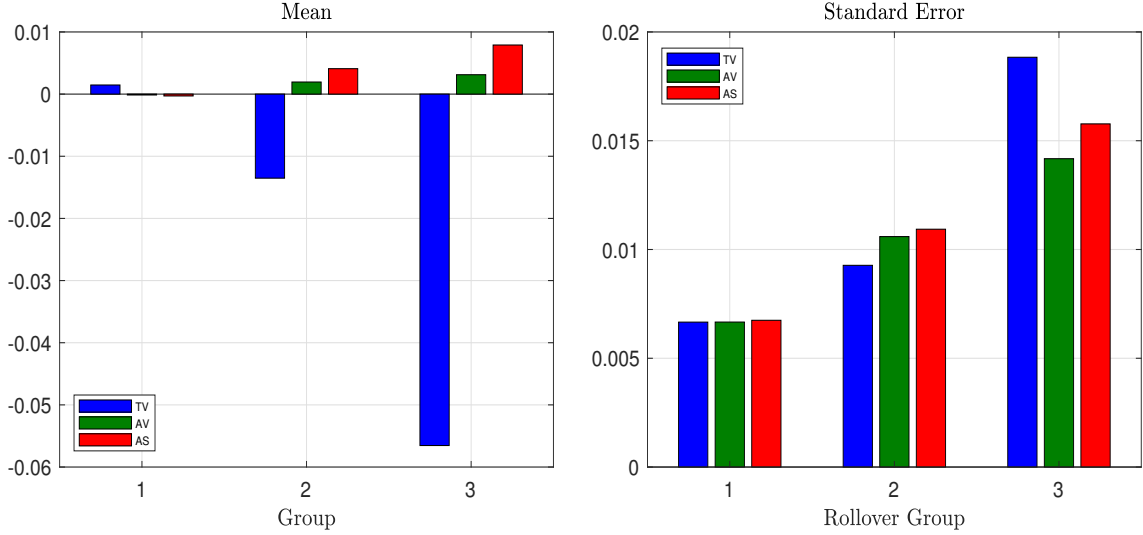
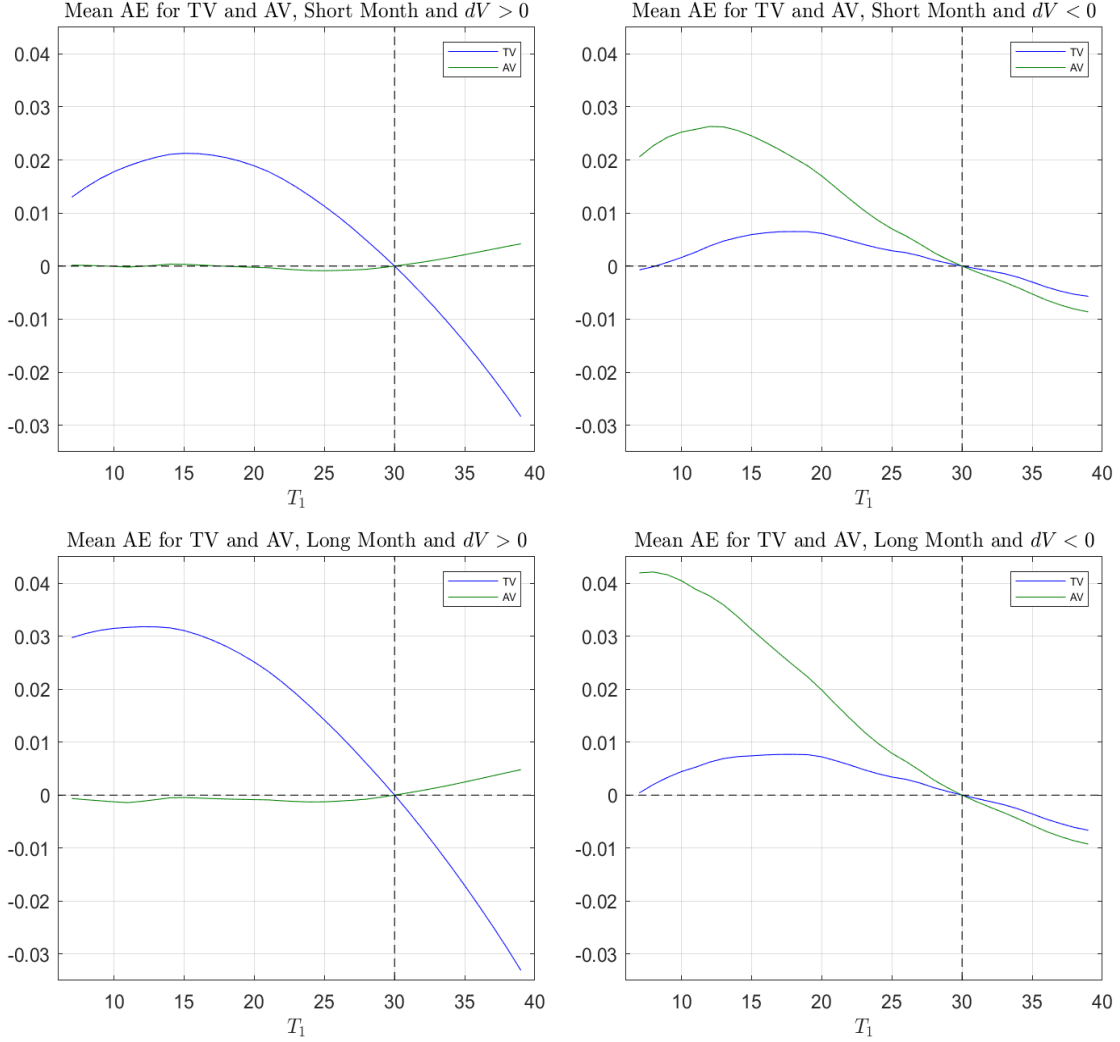


Figure 9: **Left** panel: Mean interpolation errors of  $TV_{(M)}$ ,  $AV_{(M)}$ , and  $AS_{(M)}$  volatility measures relative to VIXWE for three rollover groups. **Right** panel: Corresponding standard errors. **Group 1**: Days with no impending roll-over; **Group 2**: Roll-over dates, Short cycle; **Group 3**: Roll-over dates, Long cycle.

Due to the pronounced mean reversion of (risk-neutral) volatility, the volatility term structure tends to be mildly concave when current volatility is low, and convex when current volatility is elevated. Thus, the shape of the term structure is tightly linked to the level of volatility. Consequently, we use the slope of the term structure, measured by the difference in MFIV at the second ( $T_2$ ) versus first ( $T_1$ ) maturity date in the VIXMO computation, to proxy for the nonlinearity. The left panels of Figure 10 depict the average interpolation errors for  $TV_{(M)}$  and  $AV_{(M)}$  on trading days with an upward sloping term structure of volatility. This scenario applies for most of the trading days, providing a premium for longer-term exposures to volatility risk. The top left panel refers to errors during short maturity cycles. It shows that the  $AV$  interpolation tends to work well under these circumstances, with the  $TV$  interpolation generating substantially larger average errors. Moreover, these effects are magnified during longer maturity cycles. However, these conclusions are largely reversed in the right panels, displaying the biases during cycles with downward sloping term structures. Now the  $TV$  measures perform quite well, while the  $AV$  biases are very substantial. In fact, this is our first evidence of a severe systematic interpolation bias associated with the  $AV$  interpolation. It also provides a potential explanation for the dramatic effects during the financial crises, with elevated volatility and convex term structures, discussed in Section 4.2, where the  $AV$  measures seem particularly misguided.

To better identify the market conditions that render the  $AV$  interpolation biases significant, Figure 11 split the trading days into five quintiles, ranging from low (downward sloping) to high (upward sloping) volatility term structures. The figure provides a striking demonstration of the impact of the shape of the term structure. Whereas the left panels indicate near zero  $TV$  interpolation errors, when the term structure is strongly downward sloping, they monotonically increase as the slope increases, while the exact opposite ranking is observed in the right panels for the  $AV$  procedure. Again, these effects become stronger as the maturity cycle lengthens, as is evident from comparing the panels in the first and second rows. We conclude that the  $AV$  procedure displays lower volatility interpolation biases compared to  $TV$ , except for a dramatic reversal during scenarios involving highly elevated volatility and a downward sloping term structure. Of course, those particular days may also be the ones, where market participants and observers are especially concerned about market conditions and the evolution of volatility.



**Figure 10: Interpolation Errors for Alternative Shapes of the Volatility Term Structure.** The figure displays the average interpolation errors for  $TV_{(M)}$  and  $AV_{(M)}$  as a function of tenor and the slope of the volatility term structure. The left panels condition on a positive term structure ( $\sigma_2^2 - \sigma_1^2 > 0$ ) and the right panels condition on a negative term structure ( $\sigma_2^2 - \sigma_1^2 < 0$ ). Finally, the top row is for short maturity cycles and the bottom panels are for long cycles. The results are obtained via historical simulation over May 2012-December 2020.

For completeness, Figure 12 presents the corresponding mean absolute deviation measures for the volatility slope quintiles. They corroborate the conclusion that  $AV$  generally is superior to  $TV$ , except for the dramatic reversal in relative performance during times of volatile market conditions with steeply downward sloping volatility term structures.

#### 4.6 Summary of Findings

Our theoretical and empirical results suggest that there is a structural break in the Cboe VIX index in 2014, when the interpolation method switched from exclusive reliance on monthly options to also include weekly options with Friday expiration. Our analysis is supportive of this change in methodology, as we find the interpolation errors in the new VIXWE version negligible, whereas they occasionally are quite dramatic for the older VIXMO measure. In addition, we

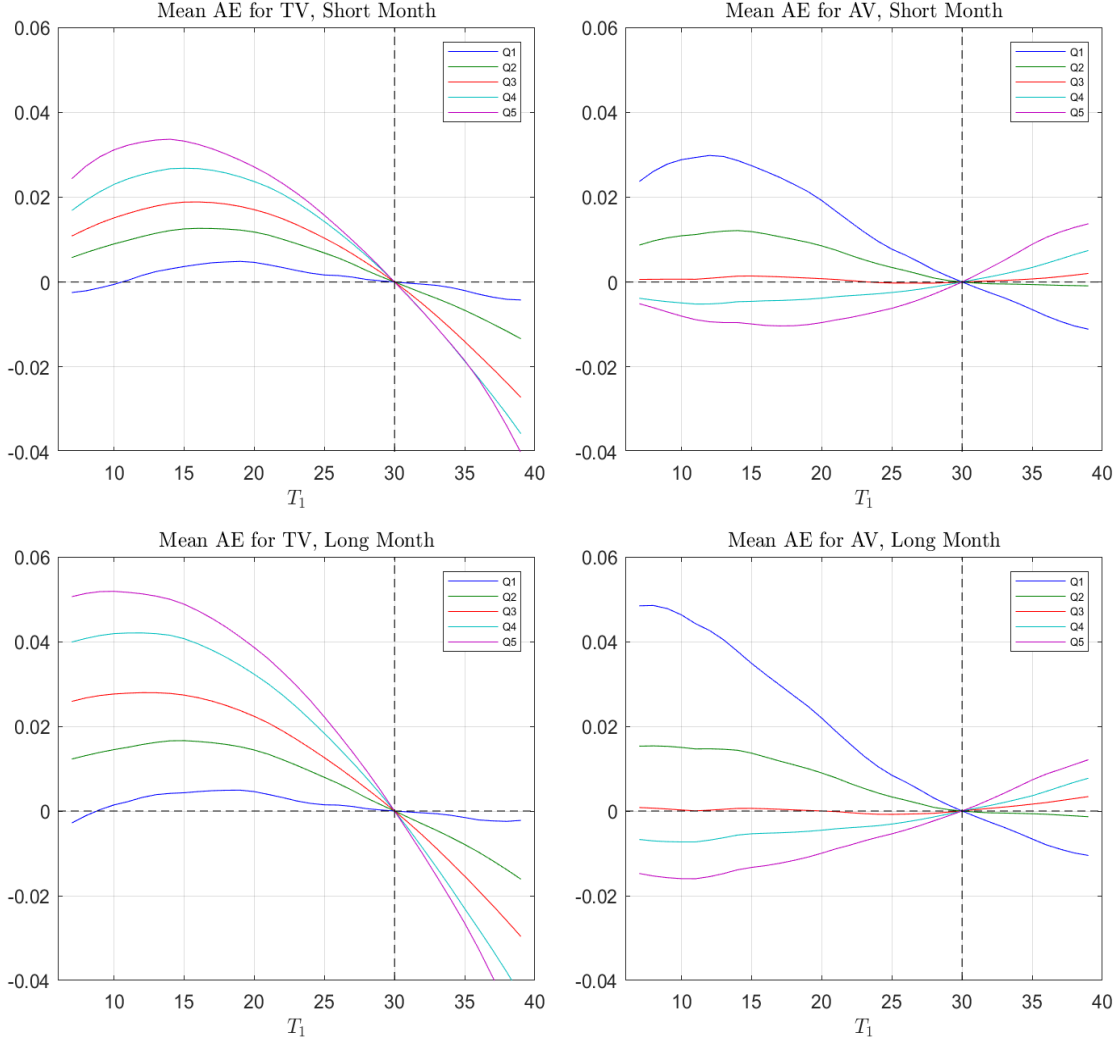


Figure 11: Simulated mean Approximation Error for  $TV$  and  $AV$  interpolations over first maturity  $T_1$ . Top panels: Short interpolation month ( $T_2 - T_1 = 28$  days). Bottom panels: Long interpolation month ( $T_2 - T_1 = 35$  days). Mean  $AE$  is shown for 5 quintiles  $Q1-Q5$  of volatility term structure ( $dV = \sigma_2^2 - \sigma_1^2$ ).

explored the potential advantages of moving from a  $TV$  interpolation procedure to an alternative  $AV$  procedure for VIXMO computation. Apart from trading days with elevated volatility levels and convex volatility term structures, we find the  $AV$  procedure to be superior relative to  $TV$ . We also concluded that, for all practical purposes, the  $AS$  approach of directly interpolating the annualized volatility, often favored in scholarly studies, is indistinguishable from  $AV$ .

The break in the VIX series is probably best illustrated through a display of our two main empirical proxy series for MFIV relative to the official Cboe VIX series over our full sample period. The top two panels in Figure 13 depict the ratio of  $AV_{(M)}$  and  $TV_{(M)}$  to VIX. The  $AV/VIX$  ratio is nicely centered on unity, but deviations of several percent on both the upside and downside are observed throughout the sample. This is to be expected, since  $AV$  interpolation differs systematically from the  $TV$ -based approach behind the VIX computation across the option maturity cycle and as a function of the volatility level. In fact, given our sense that  $AV$  is preferable to  $TV$  in most circumstances, one may argue that the  $AV_{(M)}$  measure provides a more accurate indicator of the 30-day MFIV than VIX – except for the large positive outliers,

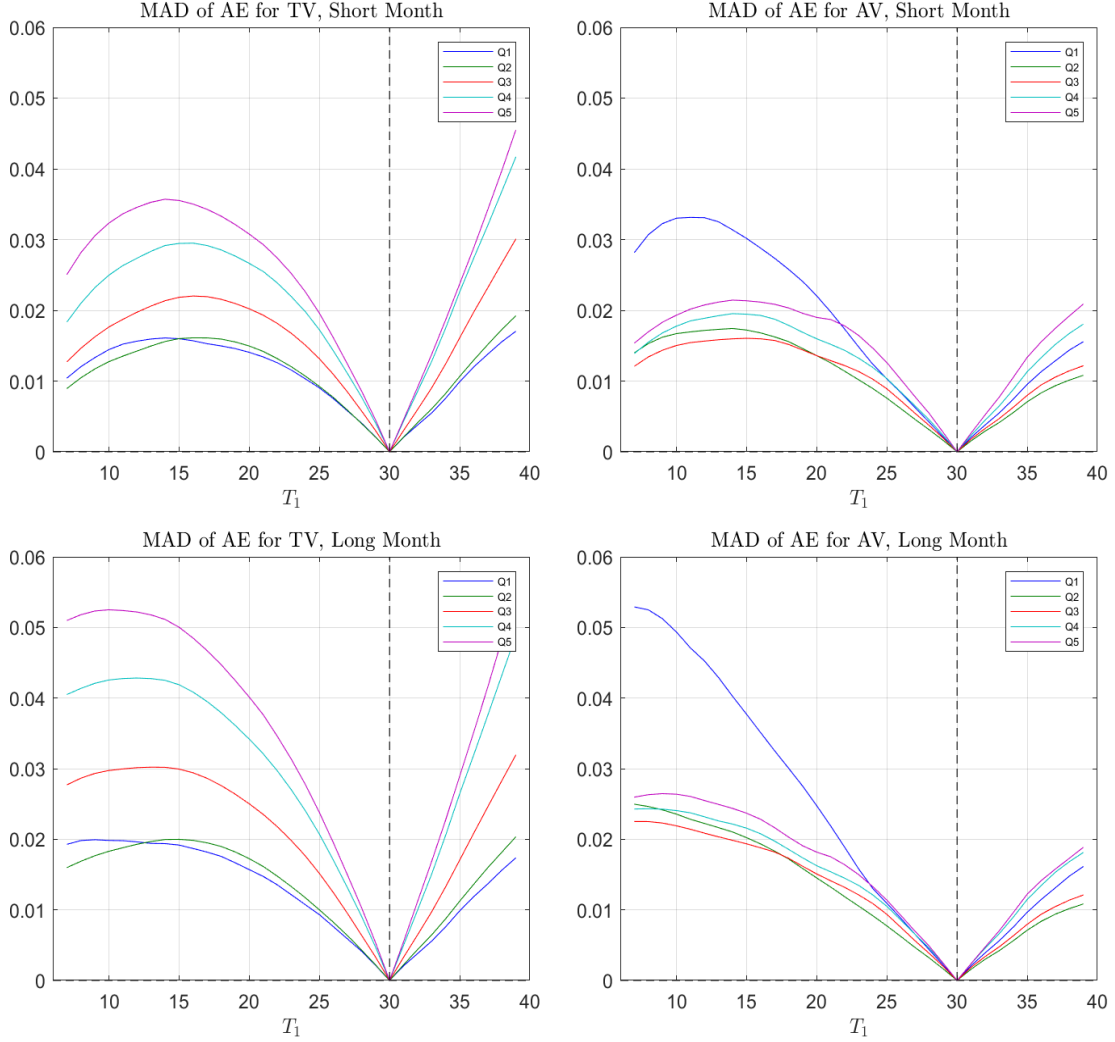


Figure 12: Simulated MAD of Approximation Error for  $TV$  and  $AV$  interpolations over first maturity  $T_1$ . Top panels: Short interpolation month ( $T_2 - T_1 = 28$  days). Bottom panels: Long interpolation month ( $T_2 - T_1 = 35$  days). MAD is shown for 5 quintiles  $Q1 - Q5$  of volatility term structure ( $dV = \sigma_2^2 - \sigma_1^2$ ).

which likely reflect the positive  $AV$  interpolation errors that can arise during turbulent market conditions. In contrast,  $TV_{(M)}$  is very close to the Cboe VIX prior to the methodological change in 2014. Again, since our  $TV$  series effectively represents an attempt at replicating the actual VIXMO procedure, this is to be expected. Of course, there are a few outliers, indicating occasional data flaws or measurement problems that are unlikely to be associated with interpolation errors.<sup>13</sup> However, following the introduction of the VIXWE methodology, the  $TV_{(M)}$  series deviates, if anything, more dramatically from VIX than  $AV_{(M)}$ . The last two panels in Figure 13 are mainly included for diagnostic purposes and to convey robustness. The third panel verifies that our two VIX proxies display similar discrepancies throughout the full sample, while the bottom panel confirms that  $AV_{(W)}$  and  $TV_{(W)}$  effectively are identical (note the difference in scale relative to the other panels), thus corroborating the claim that the interpolation errors become negligible for both our VIX proxy series, when computed from weekly options.

<sup>13</sup> Andersen, Bondarenko & Gonzalez-Perez (2015) provides a detailed analysis of the origin behind VIX replication failures.

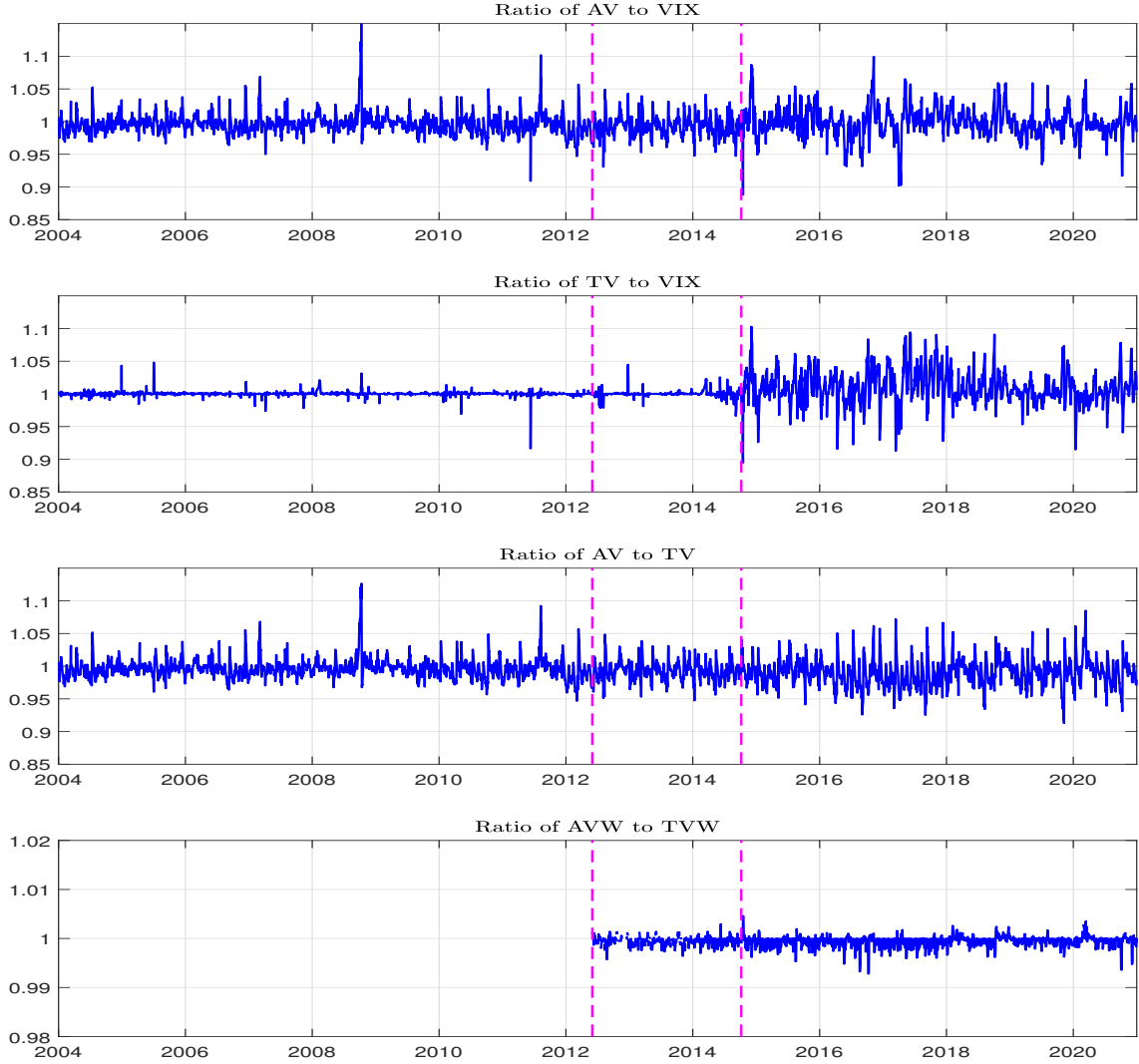


Figure 13: Various Ratios over Time.

## 5 Interpolation Errors for Alternative Tenors

Most existing VIX-style measures refer to a maturity of 30 calendar days. Of course, they may also be computed for different horizons. In fact, the partial overview in Section 2.2 references a few indices targeting the option-implied volatility for three months. Moreover, the Cboe has disseminated a 9-day and 3-month VIX for a number of years. With the rapid increase in the trading volume for short-dated options, there is growing interest in VIX-style measures covering short horizons, as evidenced by the recent introduction of a 1-day VIX. Our approach in Section 3 is readily adapted to assess the interpolation errors for alternative option maturity structures and target VIX horizons. For illustration, the following section demonstrates how our analysis carries over to the 9-day Cboe VIX measure.<sup>14</sup>

<sup>14</sup>Arguably, an even more interesting analysis would focus on the recently introduced 1-day VIX measure. However, this index is computed according to somewhat different methodology. Hence, a thorough account of the interpolation bias in that context would require a separate treatment.

## 5.1 Sources of Interpolation Error for 9-Day VIX

The 9-day VIX measure, or VIX9d, is computed using two Friday expiration options that straddle the target maturity. For example, Wednesdays will typically feature trading in Friday expiration options with about nine days to maturity, and VIX9d will be computed primarily from options with that tenor.<sup>15</sup> Ignoring Holiday closures, for the other business days, one of the two straddling Friday maturities is at most three calendar days removed from the 9-day target. As such, one may expect the associated interpolation bias to be trivial. This conjecture, however, is not only premature, but also faulty. Recall the relation  $CAF = (1 - \tau)(\tau + c)$ , along with equations (12) and (13). They imply that the maturity interpolation bias reflects both the CAF and the curvature and slope of the VIX term structure. Both features shifts in a systematic manner as we move from the 30-day to the 9-day horizon.

For CAF, the size of  $\tau = T_1/T_M$  is critical. It denotes the *relative*, not *absolute*, discrepancy between the first option maturity in the VIX computation and the target maturity. Thus, although VIXMO feature 14-day gaps between option and target maturities during each cycle, compared to 3 days for VIX9d, the relative error for the latter remains substantial. With weekly Monday and Wednesday expiry options, traded since 2016, we can readily gauge how the relative discrepancy in tenor impacts CAF. We label CAF values obtained from all available weekly options by  $CAF_d$ , and those associated with Friday expiry options only by  $CAF_w$ .

Figure 14 plots  $CAF_w$  and  $CAF_d$  during 2019, when Monday, Wednesday and Friday weeklys were traded. A number of features stand out. First,  $CAF_d$  is always smaller than (or equal to)  $CAF_w$  in absolute value, and the margin is often substantial. Second,  $CAF_w$  is always positive during regular, non-Holiday weeks. Third, the weekly option maturity cycle generates the same type of periodic pattern in  $CAF_w$  as documented for VIXMO in Figure 3.<sup>16</sup> Fourth, the low  $CAF_d$  values relative to  $CAF_w$  imply that the use of additional weekly options in the calculation of the 9-day VIX may be beneficial. However, beyond the Monday, Wednesday and Friday expiries, the gain is only marginal. For example, weekly Tuesday and Thursday expiries were introduced in 2022. Figure 15 depicts  $CAF_d$  for some non-Holiday weeks in 2017 and 2023. In the former case, Tuesday and Thursday expiries are not available, while they are in the latter. Nonetheless, the left and right panels are identical except for a spike on Tuesdays in 2017. The corresponding  $CAF_d$  value in 2023 is zero, as there is an option expiring on Thursday, 9 days later. Otherwise, the non-zero values in both panels are generated by weekdays (Thursday and Friday) where the subsequent 9-day horizon concludes during the weekend, so no option expiry is present at the target maturity, and interpolation using Friday and Monday expiries is inevitable.

For a regular business week, the maximum value of  $CAF_w$ , attained on Monday and Friday, equals  $10/81 = 0.1235$ , as illustrated by the blue periodic pattern in Figure 14. Hence, compared to the 30-day VIXMO CAF,  $CAF_{9d}$  is generally lower, albeit not by an order of magnitude. This brings us to the second key determinant of the interpolation error, namely the volatility term structure at the target maturity. It tends to be both substantially steeper in absolute value and more volatile, on average, for the 9-day than the 30-day horizon. As a result, we may expect the VIX9d interpolation errors to be as problematic as they are for VIXMO.

<sup>15</sup>Whenever a monthly and weekly option expires on the same Friday, only the AM-settled monthly option is considered for inclusion in the VIX9d computation.

<sup>16</sup>The occasional spike in  $CAF_d$  in Figure 14 arises due to breaks in the maturity cycle caused by Holiday exchange closures.

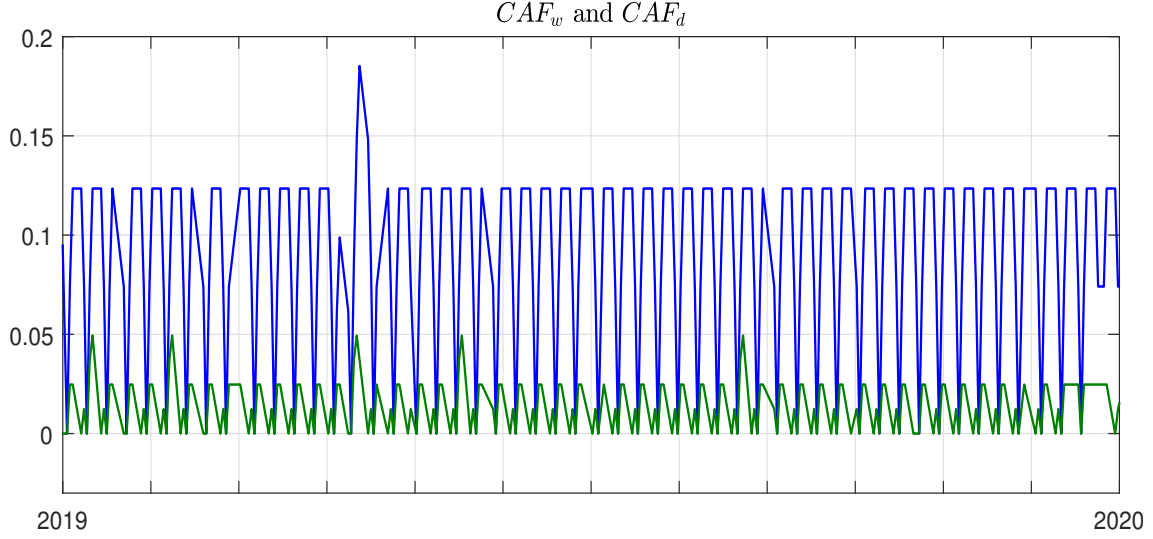


Figure 14: **Alternative VIX9d Convexity Adjustment Factors, 2019.** 9-day CAF obtained from Friday expiry options ( $CAF_w$ , blue) vs. straddling Monday, Wednesday or Friday options ( $CAF_d$ , green).

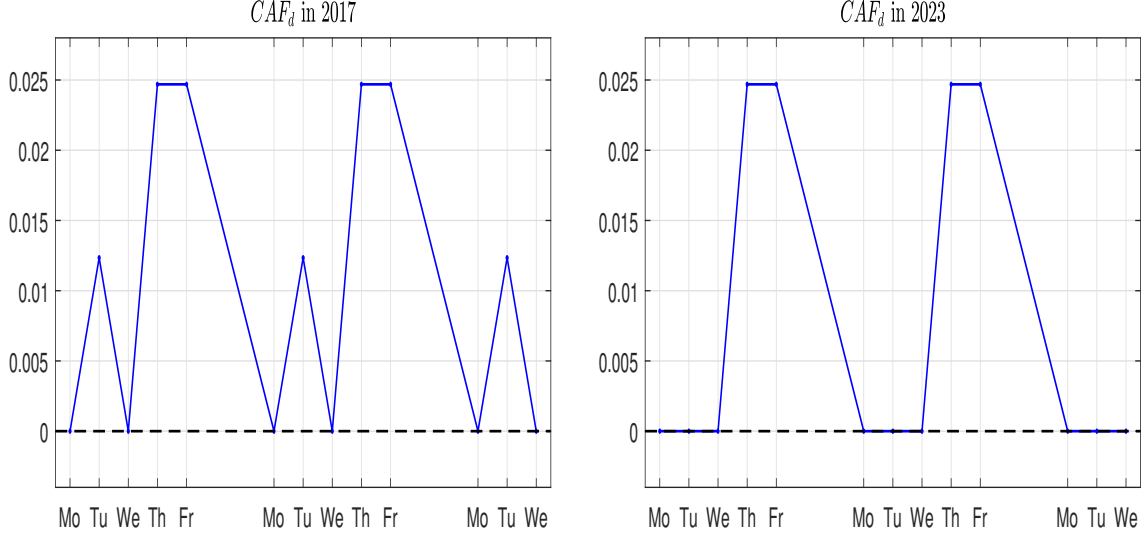


Figure 15: **VIX9d Convexity Adjustment Factor in 2017 and 2023.**

## 5.2 Empirical Analysis of 9-Day VIX Interpolation Errors

We proceed in a manner analogous to our VIXMO analysis, exploiting the fact that weekly options with Monday and Wednesday expirations have been trading, alongside the original Friday weeklys, since 2016.

We already noted that the maturity structure underlying the VIX9d induces a pronounced cyclical pattern in  $CAF_w$ , as displayed in Figure 14, reminiscent of the pattern in the 30-day measure in Figure 3. Since the corresponding  $CAF_d$  is much smaller, it should mitigate any systematic interpolation bias associated with the official VIX9d measure. For brevity, we label the 9-day VIX measures computed from Friday expiry options versus Monday, Wednesday and Friday weekly options by  $TV_w$  and  $TV_d$ , respectively. Figure 16 displays the autocorrelogram for the ratio of the official VIX9d ( $TV_w$ ) and  $TV_d$ , with the latter having the more granular expiry structure. In qualitative terms, it is strikingly similar to Figure 2, apart from the different

cycle length. Absent any systematic errors, the two indices should capture the same underlying quantity - the risk-neutral expected return variation over the future 9 calendar days. However, given the strict periodic pattern, we have a strong indication of systematic interpolation errors. Specifically, the gap from peak to peak or trough to trough is close to five business days, and the distance from peak to trough is even larger than observed for the 30-day VIX ratio in Figure 2. This is consistent with the weekly  $CAF_w$  cycle generating large interpolation errors in the official VIX9d index compared to the 9-day VIX value obtained from a broader set of weekly options,  $TV_d$ . In fact, adopting the perspective that the  $CAF_d$  values in Figure 14 are sufficiently small that the associated interpolation errors are negligible, we may interpret the deviation between these two 9-day VIX measures as an indicator of the interpolation error for the official 9-day VIX, exactly as we did earlier for VIXMO versus VIXWE.

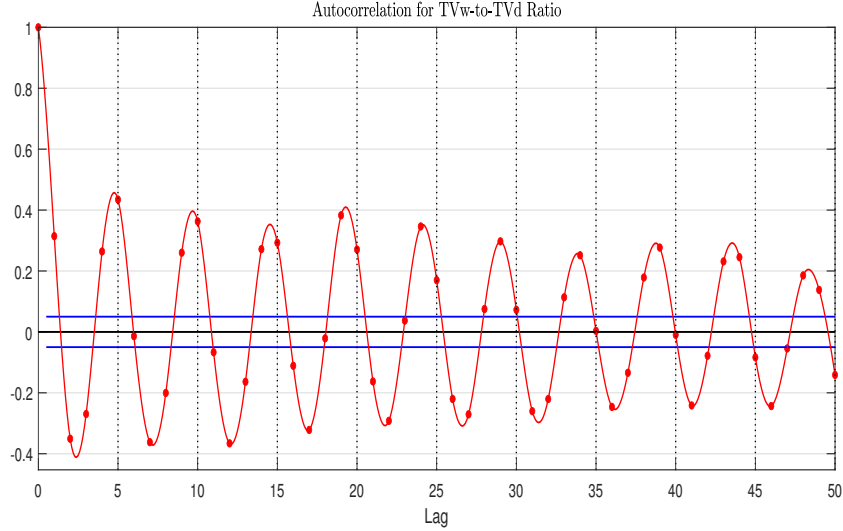


Figure 16: Autocorrelogram for the ratio of  $TV_w$ -to- $TV_d$ . The horizontal axis indicates the lag measured in exchange trading days. The horizontal blue lines represent two standard error bands under the assumption that the measurement errors are i.i.d.

Of course, there are alternative interpolation techniques, and it is desirable that our VIX9d measure is robust to different procedures. In Section 4, we found the  $TV$  and  $AV$  schemes to generate near identical VIXWE measures, and thus identical indicators for the interpolation errors. Here, the analogous task is to assess the consistency between the  $TV_d$  and  $AV_d$  9-day measures based on all available weekly options relative to the discrepancies between  $TV_w$  and  $AV_w$ , that is, the official VIX9d index computed via  $TV$  or  $AV$  interpolation. These comparisons are provided in Figure 17. The top right panel shows that the  $AV_w$ -to- $TV_w$  ratio displays non-trivial variation across our sample, with some indication of systematic deviations across different time periods. This is expected, as the two procedures are subject to different biases depending on the shape of the term structure of volatility. However, overall,  $AV_w$  is only slightly lower than  $TV_w$ , so the deviations largely balance out over time. In contrast, from the top right panel, we see that the  $AV_d$  and  $TV_d$  measures are closely aligned, with only trivial deviations across the sample. We conclude that, indeed,  $TV_d$  may be viewed as a robust 9-day VIX measure, as the associated  $CAF$  values are low and it is largely immune to the interpolation scheme.

We now turn to the question of the extent of interpolation errors embedded in the VIX9d, as manifest in its practical implementation through the  $TV_w$  measure. We note from the lower left panel in Figure 17 that  $TV_w$  generally deviates substantially from and, importantly, is strongly



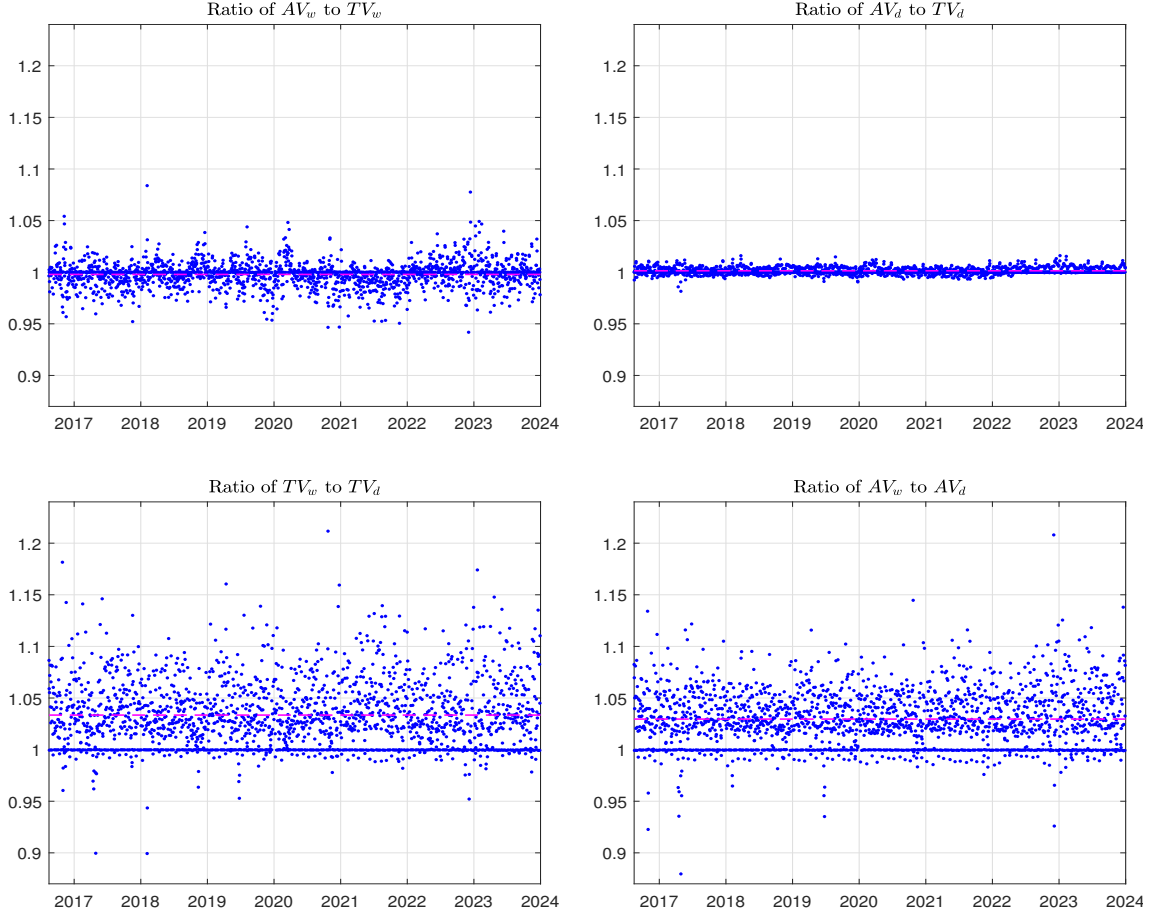


Figure 17: Time-Series of Various Ratios.

upward bias relative to  $TV_d$ . Interpreting the deviations as a percentage interpolation error, they are even larger and display more bias than for VIXMO. Furthermore, the lower right panel demonstrates that this finding is not tied to the specific interpolation method, as the  $AV$  approach generates a qualitatively identical picture. This is perhaps slightly surprising, as we can infer that  $TV_w$  and  $TV_d$  must be near identical on Wednesdays during periods not impacted by Holidays, as both rely almost exclusively on the 9-day Friday expiry options in the VIX9d computation in this scenario. To shed some light on the issue, the left panel of Figure 18 breaks down the  $TV_w$ -to- $TV_d$  ratio into weekdays. As expected, the Wednesday bias is negligible, but we see that all other weekdays display a significant upward bias for the  $TV_w$  measure.

Upon reflection, biases for all non-Wednesdays should have been anticipated.<sup>17</sup> They will naturally arise from the discrepancy in return volatility over regular weekdays versus non-business weekend days - Saturday and Sunday. By construction, this variation is captured well by the  $TV_d$  measure, whereas  $TV_w$  tends to weight weekdays more heavily than weekend days and thus induce a positive bias. For example, on Fridays, the 9-day window will typically comprise 4 weekend days and 5 regular trading days.  $TV_d$  uses an interpolation based on the 7-day Friday and 10-day Monday expiry options, with most weight assigned to the Monday expiry, which is only one day apart from the target. Thus, it weighs one option reflecting the return variation over 5 weekdays and 2 weekend days and another option capturing 6 weekdays and 4 weekend days.  $TV_w$  uses the 7- and 14-day Friday expiry options with most weight on the 7-day expiry

<sup>17</sup>Significant day-of-week effects are also documented for the 30-day VIX by Gonzalez-Perez & Guerrero (2013).

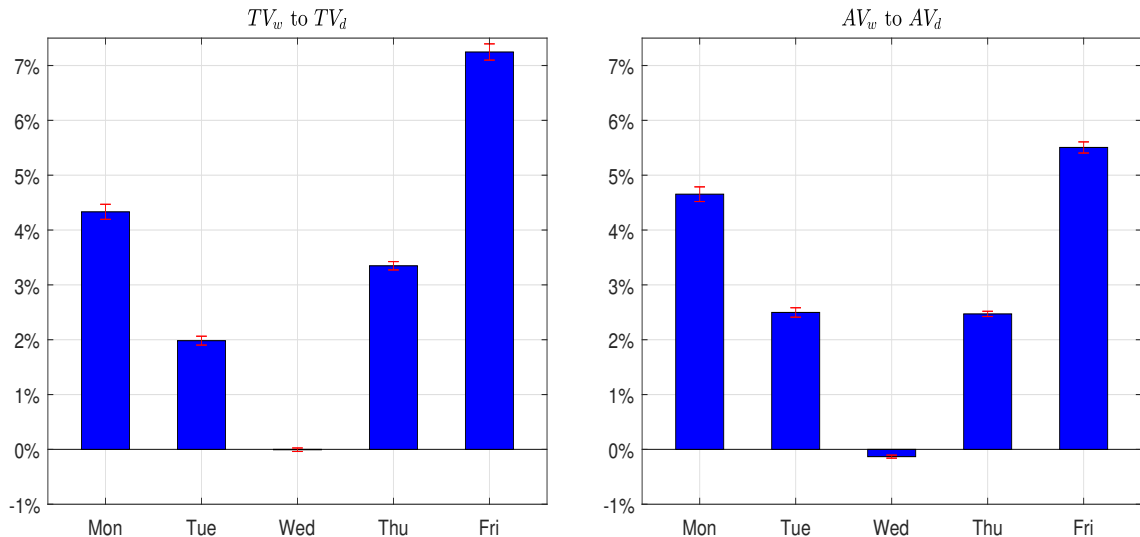


Figure 18: Mean Error of Various Ratios.

option. They reflect, respectively, 5 weekdays and 2 weekend days and then 10 weekdays and 4 weekend days. That is,  $TV_d$  rationally assigns most weight to the Monday expiry with a weekday/weekend ratio of 6-to-4 and less on the 7-day option with ratio 5-to-2. For  $TV_w$ , in contrast, the corresponding ratios for the 7-day and 14-day expiry options are identical (5-to-2 versus 10-to-4). Thus, the upward bias stems from the  $TV_w$  computational scheme overweighting the option involving a high proportion of trading days compared to  $TV_d$  (5-to-2 vs. 6-to-4). Thursdays generate a similar, but slightly less biased  $TV_w$  weight scheme.

The other weekday displaying a striking bias is Monday. On this weekday,  $TV_d$  uses the exact 9-day expiry option, involving the ratio 7-to-2. In comparison, the Friday expiry options used by  $TV_w$  feature the ratios 4-to-0 and 9-to-2. Both expiries thus capture a disproportionally large amount of trading versus non-trading day return variation relative to the proportion actually present for the 9-day horizon. The Tuesday comparison involves qualitatively identical biases, but to a milder degree. In summary, given the inherent difference in expected return variation on trading versus non-trading days, there is a valid, genuine day-of-week effect in the VIX measure. However, the use of Friday-expiry-only options to construct the official 9-day VIX exacerbates the day-of-the-week effect and induces a notable and spurious upward bias.

In summary, the argument for switching the computational procedure for the official 9-day VIX from  $TV_w$  to  $TV_d$  seems compelling. As for the 30-day VIXWE measure, it will effectively eliminate the interpolation bias. Moreover, it will largely eliminate the spurious calendar-driven upward bias in the VIX measure, although it is impossible to eradicate this entirely for long market closures around Holidays. Addressing the latter issue would require explicit ad hoc assumptions regarding the relative expected return variation over trading versus non-trading days. This topic falls outside the scope of the current investigation and is left for future research.

## 6 Conclusion

In this paper, we investigate the magnitude and significance of the maturity interpolation errors that arise from the computation of option-implied VIX-style indices. The interpolation biases depend primarily on the available option maturity structure and the term structure of volatility. We document the presence of non-trivial biases for the Cboe VIX measure up to 2014, when

the computational procedure was based on monthly options exclusively. The subsequent use of weekly options has effectively eliminated the issue as a practical concern for the 30-day VIX. Our empirical conclusions are sharpened through the adoption of a novel historical simulation technique that exploits the rich set of weekly SPX options traded actively in recent years.

Our findings should inspire other exchanges to revise their calculation procedures for VIX-style measures, as almost all other volatility indices disseminated across the globe rely on the original Cboe methodology. Moreover, we document that the interpolation biases present an even larger challenge at shorter maturities. We exemplify this through an analysis of the official 9-day VIX measure, also disseminated in real time by Cboe. This index is subject not only to random interpolation errors, but suffers from a large positive bias. That is, it systematically overstates the actual risk-neutral expected return variation. We recommend a simple change in methodology that will greatly mitigate this problem.

More generally, our analysis should facilitate and inspire the development of operational and procedural changes that improve the precision and robustness of option-implied volatility measures.

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